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- ☒ A list of figures & tables (if applicable)
- ☒ A glossary of terms (where appropriate)
- ☒ A clear statement of my project objectives
- ☒ A full reference list (the Harvard referencing style is recommended for WMG)
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Design and Evaluation of an Evidence-First Multi-Agent Workflow for UK Early-Stage Portfolio Reporting

by

Faye Niu

Dissertation submitted in partial fulfilment for the Degree of Master of Science in
Applied Artificial Intelligence

WMG
University of Warwick

Submitted _____

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

I acknowledge the use of OpenAI Codex (<https://openai.com/codex/>) as a generative artificial-intelligence tool. It was used only to support the report structure, visualisations, research and source-checking work, feedback on the writing, and technical implementation and validation. I reviewed the proposed outputs against the repository evidence and cited sources, modified or rejected them where necessary, and remain responsible for the submitted work. Appendix I records the prompts, affected work and the way outputs were checked and changed.

Project definition for my degree (as copied from the Specific Course Regulations for the course)
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This substantial individual project is designed to strengthen your academic and research skills, while allowing you to explore a specific area of artificial intelligence that reflects your interests and career goals. The project will be presented as a written dissertation, submitted at the end of the academic year. You are expected to devote an appropriate portion of your time and intellectual effort to the project during the course i.e. approximately 600 hours over the course of the academic year.
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My project relates to this definition in the following way
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My project is an individual artificial intelligence project about creating public-company research with clear supporting evidence. I designed and built a local prototype for one user and prepared a study plan. The workflow starts with a Companies House number, which a named person must check. It saves public sources, exact passages, citations and time rules before it produces claims for review. OpenAI can help find possible public URLs, but the system then collects and checks each source through its own protected process. I have not claimed a completed live company-research run or a comparison with people. The workflow keeps extraction separate from verification, records missing, conflicting and unsupported evidence, and requires a named person to approve the current version before export. The dissertation reports the prototype, study plan, checked engineering evidence, limitations and remaining work. This project therefore supports my academic and research skills in an area of artificial intelligence that matches my interests and career goals.

Abstract

Early-stage portfolio reporting combines company submissions with public evidence that may be incomplete, delayed or inconsistent. The current process has three broad stages: company information is collected, aggregated into an executive summary, and prepared for review before use. The main risks are incorrect company identity, confusion between blank and zero values, evidence from the wrong period, unresolved conflicts and claims without adequate support. This dissertation evaluates an implemented evidence-first workflow that addresses these risks through explicit data rules, traceable source records and a separately recorded verification decision. It asks how the workflow preserves identity, reporting time, missing states, sources and approval boundaries (RQ1), and what changes when a separate verification stage controls which claims enter the report (RQ2).

The prototype is a local reporting application that links each claim to its source, period, missing-data reason and approval. It separates planning, identity checks, collection, extraction, normalisation, verification, composition and approval into recorded roles with limited authority. Recorded hand-offs show who or what produced, checked and approved each statement. The roles do not imply that several agents are generally better than a simpler system. A controlled public-web route was also tested with fake providers and local fixtures, but not against a live company or website.

The evaluation used fourteen labelled fictional cases and repeated the comparison three times. Without the separate verifier, the baseline produced eleven claims: five were supported and six were not. With verification, the workflow produced the five supported claims and held all six unsupported candidates. In metric terms, precision (the share of produced claims that were supported) increased from 0.455 to 1.000, while recall (the share of required supported claims retained) remained 1.000. The combined F1 measure increased from 0.625 to 1.000, source-record completeness increased from 0.818 to 1.000, and the unsupported-claim rate fell from 0.545 to 0.000. The verifier retained an observed zero with matching evidence and held contradictory, stale and untrusted cases. These results establish the programmed mechanism on designed cases.

The contribution is therefore an inspectable evidence-first implementation together with the observed effect of its verification gate on designed cases. Because the cases and the rules were written together, the result shows that a programmed gate behaves as specified and not that the system performs well on real companies, saves analyst time or is ready for production. Manual reporting, participant review, live-web evaluation and a managed-platform comparison all lacked the observations they required, so they are set out as a prospective pilot rather than reported as findings.

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Glossary and Acronyms

This list explains the acronyms and the specialist terms used in this dissertation. Terms marked (*project term*) have a specific meaning fixed by this project’s rules rather than a general one.

Term	Meaning
Agent	A software component given one named job in a workflow, such as collecting evidence or checking a claim.
AI	Artificial intelligence.
API	Application programming interface: a fixed way for one program to ask another for data.
C0–C3	The four planned comparison conditions: manual work, a deterministic baseline, the multi-agent workflow with verification, and that workflow plus human review.
Checksum	A short fingerprint calculated from a file’s contents. If one byte changes, the fingerprint changes, so a reader can tell whether a saved file has been altered. Also called a content hash.
CLI	Command-line interface: text commands typed into a terminal rather than clicked in a browser.
Claim	(<i>project term</i>) A structured statement about one company, one measure, one value and one reporting period, which must be checked before it may appear in a report.
Contradicted	(<i>project term</i>) A verification outcome: eligible evidence disagrees with the proposed value, so the claim is withheld and the disagreement is kept on record.
Coverage	See <i>statement coverage</i> .
CSRF	Cross-site request forgery: a browser attack in which another site makes requests on the user’s behalf.
CSV, JSON, XLSX	Common file formats for tabular or structured data.
D0–D2	The three dataset tiers: D0 synthetic development cases, D1 an authorised pilot, and D2 a sealed final out-of-sample set. Only D0 produced observations.
Deterministic	Producing the same output from the same input every time, with no element of chance.
Design science	A research approach in which a working artefact is built to address a stated problem, and both the artefact and the lessons from building it are treated as the contribution.
Docker, Docker Compose	Tools that package software with its dependencies into containers so it runs the same way on different machines.
Egress	Outbound network traffic leaving a system.

Term	Meaning
Evidence item	<i>(project term)</i> A saved passage from a saved source copy, recorded with its exact location, so a later reader can find the same words again.
Fixture	A fixed, made-up test input used to check that software behaves as specified.
Held	<i>(project term)</i> A verification outcome: the claim is not supported, is out of date or comes from untrusted content, so it is withheld from the report with its reason recorded.
HITL	Human in the loop: a person must approve an output before it can be released.
HTTPS, TLS	The encrypted protocol and the encryption layer used for secure web connections.
LLM	Large language model: a model trained to generate text, such as those behind current chat assistants.
Loopback	A network address reachable only from the same machine, so the service is not exposed to the network.
Manifest	A list that binds outputs to their sources and fingerprints, so an export can be checked afterwards.
Nulls, missing states	<i>(project term)</i> The eight distinct reasons a value can be absent, such as an explicit zero, not reported, or source unavailable, kept separate rather than merged into one blank.
OOS	Out of sample: data held back so it cannot influence how a system was built.
Prompt injection	Hidden instructions placed in a web page or document that try to make a language model ignore its own rules.
Provenance	The recorded origin and handling history of a piece of data: where it came from, when it was saved and what was done to it.
RAG	Retrieval-augmented generation: retrieving documents first and then generating text from them, rather than relying on the model's memory.
RQ	Research question.
Schema	A declared structure that data must match, used to reject malformed output automatically.
Snapshot, freeze	A dated, unchanging copy of data or results, taken so later changes cannot alter what was reported.
SQLite	A small database stored as a single local file.
SSRF	Server-side request forgery: tricking a server into fetching an address it should not reach, such as an internal service.
Statement coverage	The share of a program's lines that were executed at least once by its tests. It shows what was exercised, not that the behaviour is correct.
<code>store=False</code>	A setting asking an external model provider not to save the request as application state. It does not remove separate abuse-monitoring retention.
Supported	<i>(project term)</i> A verification outcome: eligible evidence matches the proposed value, period and units, so the claim may enter the report.
Synthetic data	Made-up data written for testing. It contains no real company or personal information, and results from it describe the software, not the world.
URL	Uniform resource locator: a web address.
Verification	<i>(project term)</i> A separate check, carried out after a claim has been drafted, that compares the claim against its evidence and records an outcome without rewriting the claim.

Introduction

1.1 Early-stage portfolio reporting and company intelligence

An investor holding stakes in a dozen young companies has to say, in writing and on a date, how each one is doing. The awkward part is not writing the summary but standing behind it: a figure that came from a slide deck, a funding round that was announced but never completed, or an accounts filing that had not yet been made can each turn a confident sentence into one that cannot be defended when someone asks where it came from. The difficulty is structural rather than careless, because early-stage records are themselves incomplete and delayed (Hardman & Ramírez Santos, 2023), and a statement can carry a citation and still not be supported by it (Gao et al., 2023a). That question – where did this claim come from, and does the source actually say it – is the problem this dissertation addresses.

The verified project charter describes three reporting activities: someone gathers company information, someone turns it into an executive summary, and someone decides whether that summary may be released. Table 1.1 sets out who does what. These are jobs to be done rather than assumed job titles. The legal company and the reporting date must both be clear, because being incorporated does not prove that a business is trading, and recent records may be missing simply because a filing is not yet due (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Actor	Input	Activity and hand-off	Output	Pain point addressed
Data contributor / reporting operator	Company submission and eligible public records	Collect information and hand it to aggregation	Source records and typed observations	Mixed formats, uncertain identity, blanks and source gaps
Reporting operator	Typed observations and evidence	Aggregate and analyse, then hand a draft to review	Executive-level draft with claim links	Period mismatch, unsupported claims and hidden conflicts
Reviewer / approver	Draft, evidence, conflicts and gaps	Approve, reject or return the current version	Approved report for the downstream user	Unclear release authority and unrecorded corrections

Table 1.1. Verified portfolio-reporting process and main control point

Source note: Actor and process summary from `docs/PROJECT_CHARTER.md`. The repository evidence does not establish reporting frequency, labour cost, error rate or job title.

Evidence about a company’s development comes from dated events rather than one complete profile. Crowdfunding results vary by pitch, platform, country, sector, region and time. Published samples may

also include only successful pitches (Estrin et al., 2024). Research on repeat funding uses the campaign round as its unit and reports links with investor numbers, targets, equity offered and overfunding on one platform (Wasti et al., 2024). UK funding data can connect opportunities, applications, decisions, organisations, projects and outcomes, but its coverage and links remain incomplete (Thorne et al., 2026). A report should keep each event tied to its date and source. It should not turn these observations into a score that claims to measure company quality.

These limits show that reliable reporting starts with the data, not with well-written text. A company register may be suitable for one task but not another. Legal identity and address fields are more available than management, financial or contact information (Krasikov et al., 2020). Even required Companies House fields can be empty, so quality rules must be clear, linked to the intended use and checked in code (Nikiforova et al., 2020). Extracting financial tables can also introduce errors through optical character recognition (OCR), repeated values, cell positions and missing negative signs (Bradley et al., 2024). The system must record the source, date, location and check result for each important statement. It must also show when evidence is missing or unclear.

1.2 Problem definition and motivation

The main data problem is to match different records to the correct legal company and to use only information that was available at the relevant date. Research that joined several registers used normalised names, alternative country names, approximate matching and manual review, but some records still could not be matched (Surak & Inkley, 2026). Companies House data can contain empty required fields and inconsistent country names (Nikiforova et al., 2020). Newer companies may also have no accounts because the filing date has not arrived (Hardman & Ramírez Santos, 2023). The prototype therefore waits until the company number and legal identity are clear. It keeps different types of missing data instead of changing blanks to zeros, and records when information first became available.

Text can sound convincing and still be wrong, even when it includes citations. Research on cited answers separates factual correctness from citation precision and recall. A citation marker alone does not guarantee any of these qualities (Gao et al., 2023a). A statement may also match a source that is itself wrong (Gao et al., 2023b). Public web pages can contain hidden instructions that try to change a model’s summary, source choice or tool actions (Greshake et al., 2023). For this reason, web search is used only to find possible sources. Each report claim must point to a saved source and an exact page or passage, and a separate check must confirm the support. Retrieved text is always treated as untrusted data.

Human approval creates a release boundary, but it does not guarantee a better result. General guidance for human-AI systems recommends explaining what the system can do, showing uncertainty and allowing people to correct or dismiss its output (Amershi et al., 2019). In one controlled task, some participants followed incorrect AI advice. Prompts that made them stop and think reduced this problem, but they also affected preference, trust and different participant groups in different ways (Buçinca et al., 2021). The prototype shows conflicts, missing evidence, limits and exact source locations, and it requires a named reviewer before export. The evaluation tests the evidence and verification mechanisms, while human outcomes remain a future pilot question.

1.3 Research aim

The aim of this dissertation is to design, implement and evaluate an evidence-first, role-separated workflow for early-stage portfolio reporting. The study examines how the prototype preserves evidence from intake to a reviewable report and how a separate verification stage changes claim admission in controlled cases. This follows design-science research, which links a practical problem to the development and evaluation of an artefact (Hevner et al., 2004; Peffers et al., 2007).

1.4 Research questions

Primary research question

How can an evidence-first, role-separated workflow produce traceable early-stage portfolio reporting claims, and what effect does a separate verification stage have on claim admission in controlled cases?

The question concerns the implemented workflow rather than one model, agent label or interface. It is answered by combining RQ1 and RQ2 with the executed engineering and D0 evidence, as required in design-science evaluation (Hevner et al., 2004; Peffers et al., 2007).

RQ1: Input-to-claim accuracy and reliability

How does the implemented workflow preserve company identity, reporting time, missing states, source provenance and approval boundaries from input to report? RQ1 is answered with implementation records, engineering tests and labelled D0 cases. The assessment covers exact identity holds, value and period semantics, source-record completeness, claim-to-evidence links and repeated outcomes. These measures keep data quality tied to the reporting task and keep factual correctness separate from citation support (Gao et al., 2023a; Nikiforova et al., 2020).

RQ2: Effect of separate verification

What effect does a separately recorded verification stage have on supported-claim precision, unsupported claims, conflict handling, source-record completeness and repeatability in D0? RQ2 uses a paired comparison between C1, which has no separate verifier, and C2, which adds verification. Both conditions receive the same fictional cases and proposed values. This avoids treating a citation link or fluent statement as proof that a claim is supported (Gao et al., 2023a;b).

Scope narrowing

The original contract also proposed manual C0, named-review C3, live-web and managed-platform comparisons. They require authorised participants, frozen references, live access and recorded time, cost or review observations. Those observations were not available, so these comparisons are not active research questions. Human control may support correction and accountability, but it may also create delay or over-reliance (Amershi et al., 2019; Bućinca et al., 2021). Chapter 7 records this limitation once, and Chapters 6 and 8 specify the future pilot needed to test it.

Research objectives and evaluation plan

Four objectives connect the questions to the study. O1 defines shared rules for legal identity, dates, missing data and source records. O2 implements those rules in a workflow that stops when required evidence is missing or invalid. O3 evaluates the implemented controls and the C1/C2 verifier difference using the fixed D0 fixture. O4 distinguishes supported, unsupported, conflicting, unavailable and missing evidence in every result. Together, these objectives connect the artefact to observable evaluation criteria without treating successful execution as proof of practical benefit (Hevner et al., 2004; Peffers et al., 2007).

Table 1.2 summarises this evaluation plan and the evidence currently available.

Question	How it will be answered	Main measures	Current evidence and limit
Primary question	Combine RQ1 implementation evidence with the paired C1/C2 D0 comparison.	Traceability; decision correctness; supported-claim precision; unsupported-claim rate; source-record completeness; repeatability.	Engineering checks and D0 answer the bounded question. No real-portfolio or business-benefit inference is made.
RQ1	Inspect the implemented records and test fixed identity, date, missing-data, provenance and approval rules.	Required records and states; exact claim links; source completeness; repeated outcomes.	Repository implementation, tests and D0 provide the evidence.
RQ2	Compare C1 without a separate verifier with C2 on the same cases and with the same information access.	Precision; recall; F1; unsupported-claim rate; contradiction handling; source-record completeness; repeatability.	Executed on fourteen fictional cases across three repeats.

Table 1.2. Research questions and evaluation plan

Limit: Manual, participant, live-web and managed-platform comparisons are prospective work. Investment outcomes, production readiness and complete web coverage are not tested.

Source note: Developed by the author from the fixed evaluation plan and current evidence status. It does not add a study result.

1.5 Design alternatives and why roles were separated

A reasonable reader will ask why this problem needs several cooperating components at all. Simpler designs exist, three of them are genuinely capable, and the case for role separation is narrower than the phrase “multi-agent” usually implies. Surveys of such systems show that role labels and communication patterns vary widely and carry no guaranteed benefit (Guo et al., 2024), while design-science practice requires an artefact’s design choices to be justified against alternatives (Hevner et al., 2004). This section therefore states the case plainly, including what it does not claim.

The first alternative is a single model call: give one language model the evidence and the question, and take its answer. This is the weakest option, and not because such models write badly. The difficulty is structural. The component that writes the claim is also the component that decides the claim is supported, so there is no position from which the claim can be refused. Work on cited answers shows why that matters: whether a statement is factually right, whether its citation points to the right place, and whether the cited passage actually supports it are three separate properties, and a citation marker on its own guarantees none of them (Gao et al., 2023a). A statement can also be faithful to a source that is itself wrong (Gao et al., 2023b). If the writer is the only judge, those distinctions cannot be recorded, because nothing in the design is asking the question.

The second alternative is a deterministic script with no model at all, and here the concession is real: for structured register fields, a script is sufficient, and this project uses exactly that. The identity, period and missing-value rules in Chapter 4 are ordinary code, not model output, because fixed rules are the more reliable instrument when the input has a schema (Nikiforova et al., 2020). A script becomes insufficient only where the evidence is prose: an accounts commentary or a funding announcement carries the figure in narrative form, and extracting it involves interpretation that a schema cannot express. Table extraction alone introduces character-recognition errors, repeated values, misplaced cells and dropped negative signs (Bradley et al., 2024). The honest position is therefore a division of labour, with fixed rules wherever a rule will do and language models only where meaning must be read from text.

The third alternative is a plain retrieval-then-generation pipeline, which is the strongest competitor and the one closest to standard practice. Retrieving real passages before writing does reduce invention, and this project retrieves too. The limitation is again about authority rather than quality: in that arrangement the generator remains the last component to touch the claim, so support is asserted by the same step that composes the text, and surveys of retrieval pipelines note that retrieved passages may be irrelevant or unsupported (Gao et al., 2023c). Retrieved web content is also untrusted input, because a page can carry instructions aimed at changing a model’s summary, source choice or tool use (Greshake et al., 2023). Retrieval improves the evidence available; it does not create a stage that is allowed to say no.

What separation buys, then, is narrow and structural. The extractor cannot admit its own output, the verifier can refuse a claim but cannot rewrite it to pass, the composer can use only admitted claims, and only a named person can approve a release. These are constraints on authority, demonstrable by inspection rather than inferred from a score. The cost is equally real: more state, more hand-offs and more to maintain, and analysis of multi-agent traces shows that added roles introduce coordination and verification failures of their own (Cemri et al., 2025; Sculley et al., 2015). Managed platforms can host such roles through configuration rather than code, which Chapter 2 examines as a serious alternative (Dibia et al., 2024).

This leads to a claim deliberately smaller than the design might suggest, and it is worth stating before

any result appears. A single program could apply the identical support rule and reach the identical decisions; the D0 comparison in Chapter 5 isolates that rule and its authority over composition, not the number of components. Where Chapters 4 and 6 concede this, they state the argument’s actual shape rather than retreat from it. What is defended here is that separating who may propose, judge and approve makes the reasons for a decision recoverable – a property of the arrangement, not a measured gain in accuracy (Guo et al., 2024; Hevner et al., 2004; Wu et al., 2024).

1.6 Scope, contribution, and exclusions

The system is a local, single-user research prototype rather than a deployed service, reachable only from the same computer. Stating its purpose, user and operating limits matters because risk depends on the setting in which a system is used (National Institute of Standards and Technology, 2023). Public company information may be sent to an external model only after the required checks, while restricted or internal material stays local and is never sent to public generative-AI tools (Central Digital and Data Office, 2023).

The contribution has three parts: an evidence-first prototype, an executed controlled evaluation and a reporting method that traces claims to sources. It combines design with tests against stated objectives, without treating implementation as proof of operational usefulness (Hevner et al., 2004; Peffers et al., 2007). It also keeps factual correctness separate from citation support, so each sourced claim still needs a page or passage check (Gao et al., 2023a;b). The public-web research route provides additional implementation and adversarial-test evidence, but it is not used to answer the active questions.

The project excludes buy or sell recommendations, speculative valuations, profiles of people, automatic publication and claims that the system is ready for remote, multi-user or production use. Financial communications can affect investment decisions and may create further duties. This supports a cautious scope, not a claim of legal compliance (Financial Conduct Authority, 2024). Public web content remains untrusted because hidden prompt injection can change model output or tool actions (Greshake et al., 2023). A named person must approve a report before export, which keeps human responsibility clear (Central Digital and Data Office, 2023). These controls define the prototype’s authority boundary; their wider limits are addressed in Chapter 7.

1.7 Dissertation roadmap

Chapter 2 examines the literature behind the design criteria and weighs the architecture alternatives introduced in Section 1.5. Chapter 3 defines the executed evaluation and places unrun protocols in the appendices. Chapter 4 explains the implemented architecture and operational roles. Chapter 5 reports engineering and D0 results, including individual claim decisions. Chapter 6 interprets RQ1 and RQ2 and specifies a prospective business pilot. Chapter 7 is the authoritative location for ethics, governance and study limitations. Chapter 8 states the bounded contribution and future evidence. This order follows design science by linking the problem, artefact, evaluation and communication (Hevner et al., 2004; Peffers et al., 2007).

Literature Review

2.1 Scope and structure of the literature review

The literature relevant to this dissertation spans several connected fields rather than one established research stream. Studies of early-stage finance examine how funding rounds, public support and investor activity may signal company development, but their findings depend on the platform, sample and reporting context (Estrin et al., 2024; Wasti et al., 2024). Research on company registers also shows that public records can support analysis while remaining incomplete, delayed or difficult to link across sources (Hardman & Ramírez Santos, 2023; Thorne et al., 2026). These findings make legal identity, reporting date and source context central to reliable portfolio reporting.

A second body of work considers how raw information becomes a defensible report, treating data quality as dependent on intended use, requiring that origins and assumptions be documented, and separating factual correctness from citation support (Gao et al., 2023a; Gebru et al., 2021; Nikiforova et al., 2020). A third concerns AI-assisted retrieval, multi-agent workflows and human control, where each proposed remedy carries a matching risk: retrieval can return irrelevant or hostile material, extra agent roles add their own failures, and human review may not correct what it is shown (Buçinca et al., 2021; Gao et al., 2023c; Guo et al., 2024). The sections that follow work through these themes in turn, and the chapter closes by identifying the gap this dissertation addresses.

2.2 Early-stage finance, portfolio reporting, and public company intelligence

Founders usually know more about an early-stage business than outside investors, and funding platforms shape what information is visible. Two studies illustrate how differently that visibility can be read. Estrin and colleagues work across several countries and find that amounts raised and investor participation vary by pitch, sector, region, country and platform, but their sample contains successful pitches rather than every attempt (Estrin et al., 2024). Wasti and colleagues go the other way, studying repeat funding in depth within a single platform and taking the campaign round as the unit of analysis (Wasti et al., 2024). The two are not in conflict about facts, yet they pull in opposite directions on what a funding event means: breadth across platforms buys generalisability at the cost of a censored sample, while depth within one platform buys a clean unit at the cost of transferability. Neither design can license a general inference from investor numbers to company quality, which is why this project treats a funding round as a dated, sourced event rather than a signal to be scored.

Portfolio reports should record funding and public support as dated events with clear sources. In the Crowdcube study, an earlier successful round may act as a signal, but its meaning still depends on the campaign and platform (Wasti et al., 2024). UK research on public support found later differences between participating firms, but it also noted selection effects and did not claim that the support caused those differences (Mondal & Mellor, 2025). Rebuilding the UK funding process requires links between opportunities, applications, decisions, projects, organisations and outcomes, even though these records often lack shared identifiers (Thorne et al., 2026). A round, grant or award can support a dated fact. It cannot prove future success or provide a complete funding history.

These events need to be linked to the correct legal company, and the literature on the UK register disagrees productively about how much that register can carry. Galanakis and colleagues treat incorporation records as a timely and broad indicator of company creation, valuing the register precisely for its currency (Galanakis & Savagar, 2026). Hardman reaches an almost opposite conclusion about recent data, showing that filing rules and delays make the newest records the least complete part of the register (Hardman & Ramírez Santos, 2023). Krasikov and colleagues then add a third position: open access is not the same as readiness, because a register may be fully published and still unsuitable for a given business task (Krasikov et al., 2020). The disagreement is real but resolvable, and the resolution is that timeliness holds for the fact of incorporation and fails for financial filings. This project therefore accepts register identity as authoritative while treating recency of financial records as a property to be checked per field rather than assumed.

Two constraints follow for what may be reported. Public-support differences may reflect which firms applied rather than any effect of the support (Mondal & Mellor, 2025), and financial communications must present benefits and risks fairly (Financial Conduct Authority, 2024). Together these rule out causal performance claims, speculative valuations and any general company-quality score.

2.2.1 Sector importance and existing reporting approaches

The reporting problem sits within a material but uneven early-stage investment market. The British Business Bank records £10.8 billion of announced UK smaller-business equity investment across 2,048 deals in 2024, while deal numbers fell by 15.1 per cent and seed- and venture-stage activity weakened (British Business Bank, 2025). Monitoring also consumes professional attention: a survey of 885 institutional venture capitalists reported an average of 18 hours each week working with portfolio companies and about 60 per cent interacting with them at least weekly (Gompers et al., 2016). These descriptive results come from different periods and settings; monitoring is broader than reporting, and neither source measures the effect of a reporting tool.

Private-market evidence also limits what reporting can consolidate. The British Business Bank treats angel-investment evidence as partial because much activity is private and reporting is patchy (British Business Bank, 2025), and register research reaches a parallel conclusion about public records being published yet unfit for a given task (Krasikov et al., 2020). Portfolio managers, analysts and downstream report users therefore need dates, definitions, legal identities and visible coverage gaps. Taken together, these sources establish the reporting problem as one of traceability under incomplete evidence, which is a different problem from the efficiency case usually made for automation.

Several technical approaches are plausible. Manual documents and spreadsheets preserve flexible judgement, but their provenance and consistency depend on the local practice. Deterministic data pipelines can apply repeatable identity, time and quality rules, although lineage must be explicit and schema changes can invalidate those rules (Buneman et al., 2001; Nikiforova et al., 2020). Retrieval-augmented

generation can extend unstructured search but may retrieve irrelevant or hostile material, while one fixed process or several bounded roles can apply the same final support rule (Gao et al., 2023c; Guo et al., 2024). The reviewed sources do not measure this project’s manual workflow, so effort and error cannot be assigned to any approach before a matched study.

Commercial products cover related but non-equivalent workflows, and setting them beside the data-quality literature exposes an unresolved tension at the centre of this field. Visible describes portfolio-data collection, monitoring, dashboards and investor reporting; Zapflow describes integrated deal and portfolio records; and Affinity describes relationship-centred activity capture and reports (Affinity, n.d.; Visible, n.d.; Zapflow, n.d.). Each presents consolidated portfolio figures as a solved presentation problem. Yet Kaplan and Lerner describe the underlying private-market data as incomplete and inconsistent, with definitional differences, missing failed firms and backfilling (Kaplan & Lerner, 2016). Both cannot be straightforwardly true: a dashboard cannot be more reliable than the records beneath it, and no vendor page reports claim-level support accuracy or how conflicting sources are resolved. That gap between confident presentation and acknowledged data weakness is the space this dissertation works in. Table 2.2 compares documented functions and unresolved evidence rather than scoring products.

2.3 Data quality, company identity, missing data and time

Data quality is a property of a dataset in use, not of the dataset alone, so accuracy, timeliness, completeness and consistency each need rules bound to a particular field and purpose (Krasikov et al., 2020; Nikiforova et al., 2020). The practical consequence is that a portfolio report must state how each field will be used and which checks matter for it, because a single combined quality score can conceal a serious fault in one field.

Company matching also depends on the source. An exact company number from the official register is the main legal identifier; a company name is only a possible match. Funding sources without shared identifiers need careful exact or approximate matching, and errors and gaps still remain (Thorne et al., 2026). Work across several registers also required name normalisation, approximate comparison and manual checks when company or country names disagreed (Surak & Inkley, 2026). Companies House identifies incorporated companies rather than all economic activity (Galanakis & Savagar, 2026). An uncertain match should therefore stay unresolved until a named reviewer checks it. Records with similar names must not be merged automatically.

Missing data must keep its meaning instead of becoming a default number. Documentation should state what is missing, why it is missing and what is still unknown (Gebru et al., 2021). A financial table contains values, empty cells, positions and different types of information. An empty cell does not automatically mean zero (Bradley et al., 2024). Completeness and validity are also separate qualities that need their own checks (Nikiforova et al., 2020). This project distinguishes a blank, zero, none, not applicable, not reported, not found publicly, invalid and observed value. Not found publicly does not mean that something does not exist. These categories are a project design choice, not a general standard from the literature.

The system also needs separate dates. It records when an event happened, when the information became available, when it was collected and the last date allowed in the report. Register snapshots and filing delays show why recent facts may not have been available at an earlier reporting date (Hardman & Ramírez Santos, 2023). Incorporation and later business activity also happen at different times (Galanakis & Savagar, 2026). Using information published after the reporting date creates data

leakage and may make an evaluation look better than it should (Kapoor & Narayanan, 2023). A correction needs a new version and date. It must not be used to support an earlier report.

Together these rules give a testable evidence model, recording company identity, the type of missing value, event and availability dates, source, exact passage and version, and keeping invalid, conflicting and unresolved records visible. Its checks are tied to the task and the data (Gebru et al., 2021; Nikiforova et al., 2020), and its dating rules exist to keep later information out of an earlier report (Kapoor & Narayanan, 2023). Chapter 3 puts this model to work.

2.4 Source records, source checks, conflicts and traceable reporting

Provenance is the record of where a statement came from and what was done to it. Foundational database work separates why a query result exists from where its contributing values sat (Buneman et al., 2001); that explains lineage, not whether a report is right. Documentation should record purpose, contents, processing, gaps, errors, dependencies and corrections (Gebru et al., 2021), and repeating a result needs enough data, code, steps and measures (Pineau et al., 2021). Each portfolio statement should therefore link to its saved source, its exact place in that source, the changes made to it, the checks applied and the approval given.

Checking a source and checking a claim are two different steps: the source check covers access, type, permission, date and whether a stable copy can be saved, while the claim check asks whether an exact passage supports the statement. Two studies by overlapping authors take notably different views of what to do when that second check fails. The ALCE work treats attribution as a property to be measured, separating factual correctness from citation precision and recall so that weak support becomes visible in the score (Gao et al., 2023a). The RARR work treats it instead as a defect to be repaired, revising generated text after the fact so that it matches retrieved evidence (Gao et al., 2023b). Both are defensible, and the choice between them is consequential rather than technical: repair improves the finished text but relocates the decision back inside the component that produced it. This project follows the measurement position and refuses repair, holding an unsupported claim rather than editing it into compliance, with the decision and files recorded so the outcome can be repeated (Pineau et al., 2021).

Where permission and technology allow, an accepted source should be saved as a fixed local copy with a checksum, collection time, exact passage and version, and a correction should create a new version rather than replace the old one. Dataset and reproducibility guidance both support recording external resources, updates, errors and superseded versions (Gebru et al., 2021; Pineau et al., 2021), and model cards apply the same logic to versioned intended use and evaluation limits (Mitchell et al., 2019). Some sources, of course, cannot legally or technically be saved at all.

Conflicting evidence must be retained. Hallucination research distinguishes factual conflict, unverifiable claims and overstatement, linking reliable output to uncertainty and checking against trusted knowledge (Huang et al., 2023). A source link does not resolve disagreement when text conflicts with evidence or the source is wrong (Gao et al., 2023b); nor can one citation marker represent correctness, passage support and completeness (Gao et al., 2023a). Supported disagreements remain grouped and unverifiable claims unresolved until a stated rule or named reviewer decides them. The system must not automatically prefer the majority, newest or most fluent version.

A traceable report therefore records the whole chain: discovery, checking, capture and use, then the

claim, support decision, conflict, approval and export. The design principle drawn from this section is that a verifier should check evidence without being able to rewrite the claim, precisely because correctness, support and attribution are distinct properties (Amershi et al., 2019; Gao et al., 2023a;b; National Institute of Standards and Technology, 2023). Figure 2.1 summarises this admission chain. The supporting review-source, signal-boundary and data-semantics tables are provided as Tables H.1, H.2 and H.3 in Appendix H.

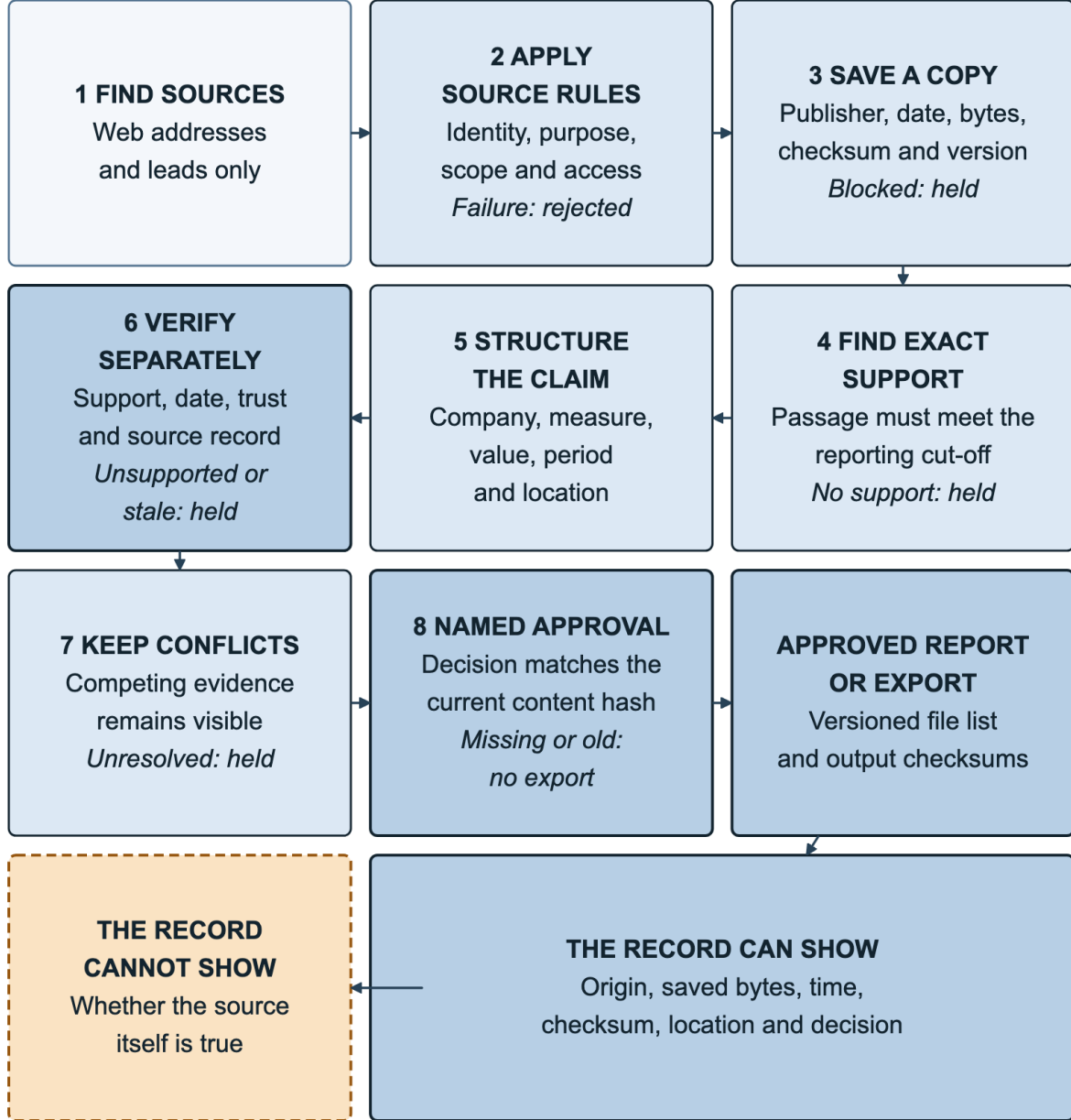


Figure 2.1. How a possible source becomes an approved report claim. Developed by the author from the cited literature and the project’s fixed rules.

2.5 LLM-assisted source finding and evidence extraction

Retrieval-augmented generation (RAG) separates document indexing, retrieval and text generation, and the literature holds two views of it that are hard to reconcile. In the hallucination literature,

retrieval is the remedy: grounding output in real documents reduces invented content that a model would otherwise produce from memory (Huang et al., 2023). In the security literature, the same mechanism is the vulnerability, because indirect prompt injection lets text hidden in retrieved pages alter a model’s summary, queries, source choices or subsequent tool use (Greshake et al., 2023). Retrieval, in other words, fixes a problem of invention by opening a channel for a problem of influence, and surveys of the field note that retrieval may equally miss relevant material or return irrelevant passages (Gao et al., 2023c). The practical consequence adopted here is that a large language model (LLM) may help find candidate sources, while finding, checking, extracting and verifying stay separate steps that retrieved text cannot reach into.

The saved Responses API documentation describes model-managed `web_search`, inline citations, annotations, domain filters and a list of consulted sources (OpenAI, n.d.-b). These features can help plan a search and produce a list of possible sources (Gao et al., 2023c). They guarantee nothing about how many sources or citations come back, though, and a visible citation may still be incomplete or unsupported (Gao et al., 2023a). Returned URLs, snippets and model text stay possible sources until they pass the source checks.

Structured evidence should first be extracted with fixed, rule-based code. Unstructured public or synthetic evidence may use an optional model, but only to suggest a claim that links to an exact passage. Factual correctness, passage support and citation completeness remain different measures (Gao et al., 2023a). A correct source link also does not prove a claim when the evidence disagrees (Gao et al., 2023b). Both routes end with fixed checks of the exact passage, data structure and format, followed by separate verification. This reflects the need to test relevance, support, rejection and reliability separately (Gao et al., 2023c). Unsupported content is rejected or left unresolved. It is never filled in from model memory.

Public content found through search must be treated as untrusted data, not as instructions. Indirect prompt injection can hide harmful instructions in searched material and change summaries, queries, source choices or later actions (Greshake et al., 2023). RAG can also receive noisy, conflicting or hostile text (Gao et al., 2023c). The prototype therefore limits tool permissions, checks each destination, separates extraction from other actions, limits redirects and requests, and stops safely when a check fails. Testing, uncertainty, separate review and safe failure are recognised risk controls (National Institute of Standards and Technology, 2023).

A report should therefore state how much searching was permitted rather than imply that the web was searched. Context size, token limits, domain filters and source lists are all configurable, and the number of sources returned is not guaranteed (OpenAI, n.d.-b), so a repeatable method records the queries, sources, reporting date and budgets, then names what was inaccessible, missing, conflicting or unverified (Gao et al., 2023a; Pineau et al., 2021).

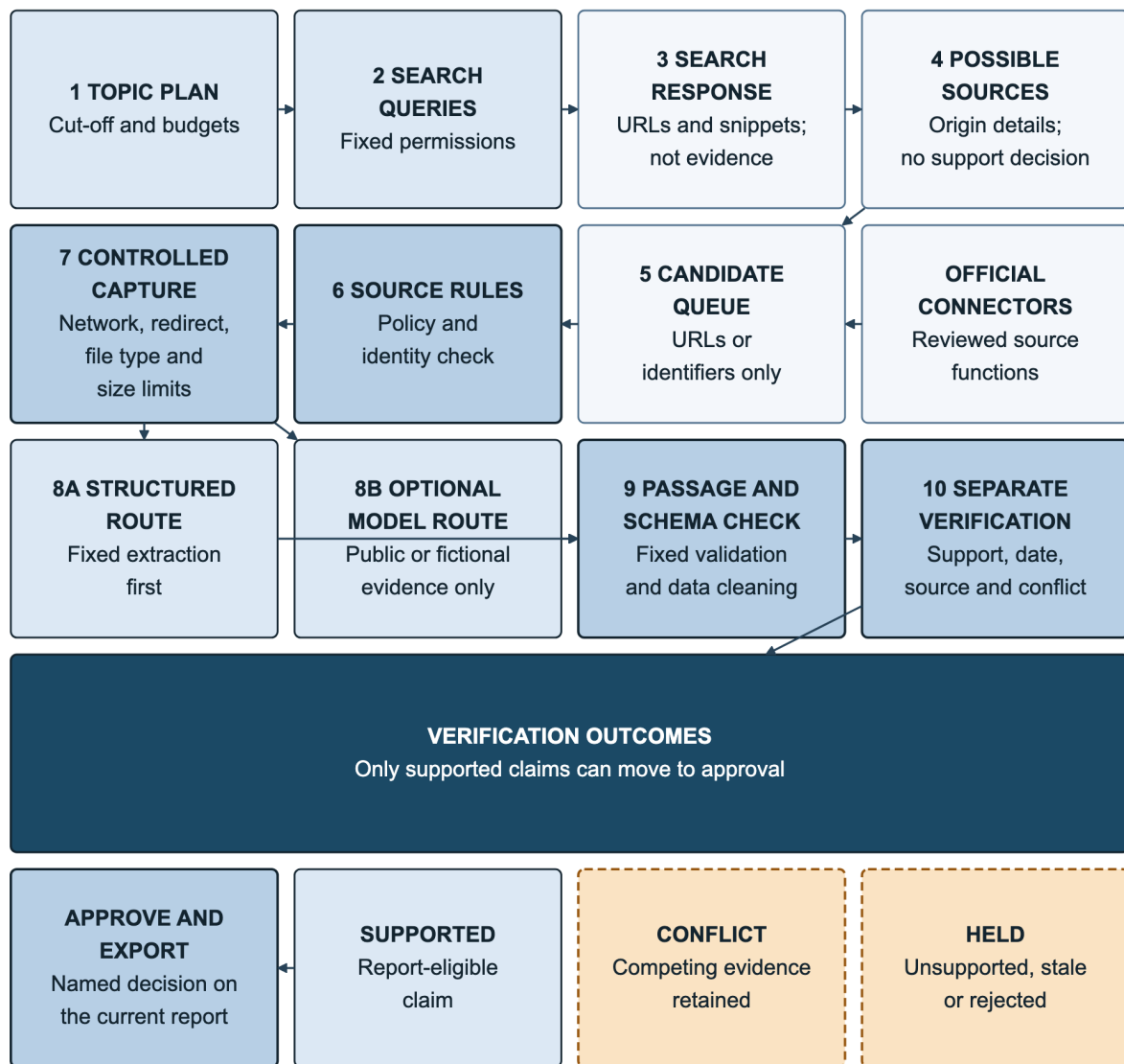


Figure 2.2. Why finding a source is not the same as having evidence. Developed by the author from the cited literature and the project's fixed rules.

2.6 Functional role separation and human approval

Dividing a task between programmed roles is a design choice, not an acquisition of collective intelligence, and the literature is strikingly inconsistent about what such a division even is: reported systems differ in interface, role description and communication structure (Guo et al., 2024), while AutoGen shows that the developer, not the architecture, fixes the roles, tools, control flow and stopping rules (Wu et al., 2024). This is why Table 2.1 assigns each role one responsibility, defined inputs and outputs and a visible failure limit, and why Section 1.5 rests its case on authority rather than on role count (Peppers et al., 2007).

The clearest counterweight to any enthusiasm here is empirical. Analysis of 1,642 traces from seven multi-agent systems identified fourteen failure modes spanning system design, inter-agent alignment and task verification (Cemri et al., 2025). That evidence deserves careful reading rather than either dismissal or alarm: much of the wider dataset was annotated with model assistance, and its coding and mathematics tasks differ from portfolio reporting. Read cautiously, it still establishes that added roles are themselves a source of failure, which makes programmed control flow, tool permissions and explicit stop rules necessary rather than optional (National Institute of Standards and Technology, 2023; Wu et al., 2024).

Human review gives a person authority to decide; it does not create new evidence. Here the applied guidance and the experimental evidence are in genuine tension. The widely cited design guidelines recommend making system limits clear, showing uncertainty and letting people dismiss or correct output, treating transparency as the route to sound human judgement (Amershi et al., 2019). Bućinca and colleagues tested something close to that prescription and found it insufficient: participants still followed incorrect advice, and the interventions that reduced over-reliance did so by making the task deliberately harder, with uneven effects across participant groups (Bućinca et al., 2021). The uncomfortable implication is that the remedy the guidelines propose is the one the experiment found wanting, so a well-designed review screen cannot be assumed to produce better decisions. This project accordingly treats its interface as an accountability mechanism rather than an accuracy mechanism: it shows the claim, exact support, conflicts, uncertainty and content hash so that a named reviewer’s decision is recorded and attributable, and it leaves the question of whether review improves outcomes to a study with participants (National Institute of Standards and Technology, 2023).

Evaluating such a workflow therefore means keeping its failures visible rather than reporting only its successes. Logs should carry role, task, attempt, input, output, hash, time, tokens, errors and recovery, and fixed cases should cover each role and the whole workflow, including disagreement and safe failure (Guo et al., 2024; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Managed platforms make custom orchestration itself look optional, which is the sharpest challenge to this project’s design. AutoGen Studio already exposes models, tools, memory and workflows through a visual interface, so much of the wiring described above can be bought rather than written; its authors nonetheless call the tool not production-ready and name architecture comparison and reliability as open problems (Dibia et al., 2024). Vendor documentation for Grok Bot goes further, describing persistent named Bots with distinct jobs, tools, working rules and approval boundaries (SpaceXAI, 2026). The unresolved question is not whether a platform can host roles, but whether it can enforce an evidence contract; convenience in configuring agents is a different property from correctness in admitting claims, and neither source measures the second (Dibia et al., 2024; Guo et al., 2024).

Role	Permitted input / tools	Required output	Stop / failure condition	Forbidden authority
Query planner / discovery	Fixed question, cut-off, source rules and limits; approved search or catalogue interfaces.	Structured search plan and possible-source queue with origin, scope and stop reason. Possible sources are suggestions only.	Scope, time, attempt or tool limit reached; no eligible candidate; policy mismatch. Record empty, missing or held state.	Accept a source; treat a suggestion as evidence; create a report claim; widen permissions or approve actions.
Evidence extractor	Accepted write-once evidence, exact location and structured schema; fixed parser or approved extractor with clear limits.	Structured extraction proposal with exact passage, value or state, date and source record.	No exact support; invalid schema; ambiguity, conflict, stale evidence, budget or tool failure. Abstain or hold.	Fill gaps from memory; change source rules; hide missing information; decide support, report eligibility or approval.
Independent verifier	Producer record, write-once evidence or passage, source record and fixed support rules.	Separate assessment recording supported, contradicted, stale, insufficient or rejected-untrusted, with reason.	Evidence or location absent; value, period, trust or source mismatch; unresolved conflict or rule failure. Reject or hold.	Silently rewrite the producer record; seek replacement support; override policy; compose, approve or export.
Fixed report writer	Verified eligible claims, clear exceptions, agreed template, ordering and citation rules.	Repeatable draft report with citations, conflicts, unknowns, limits and content hash.	Any claim lacks an eligible verification outcome; template, schema or hash invariant fails. Refuse composition.	Add or infer facts; repair missing evidence; alter verification; hide exceptions; approve or publish.
Named human reviewer	Versioned report, exact support, verifier records, conflicts, uncertainty, limitations and content hash.	Named, timed approve, reject or return-for-correction decision with rationale and report version.	Identity or version mismatch; unresolved support or conflict; incomplete evidence; no review authority. Do not approve.	Manufacture support; silently change the approved version; override source eligibility; treat review as proof of truth or utility.

Table 2.1. What each workflow role may and may not do

Meaning: Developed by the author and not a result. Programmed roles and messages do not show independent intelligence, better performance or benefit.

Authority rule: Model-assisted roles may suggest or assess records within set limits. The verifier cannot quietly rewrite a claim, the report writer cannot add facts, and the reviewer adds authority to decide rather than new evidence.

Source note: The local sources are guo2024multiagent, wu2024autogen, peffers2007dsrm, nist2023airmf, amer-shi2019guidelines, bucinca2021forcing and pineau2021reproducibility. Supporting pages are in claim-ledger rows 2.6-P1–2.6-P4.

2.7 Main findings from the literature and research gap

No single best reporting design emerges from this literature, and the disagreements traced above explain why. Private-market datasets have limits of coverage and definition, work on lineage and citation keeps a statement’s origin apart from its support and its truth, and vendor pages offer no independent comparison (Buneman et al., 2001; Gao et al., 2023a; Kaplan & Lerner, 2016; Visible, n.d.). What is missing is therefore evidence rather than novelty: whether rules for identity, missing values, time, source records, verification and human authority hold together on the same early-stage inputs.

For RQ1, the established problem is that company and funding records can be incomplete, delayed and hard to link, while an exact citation can still sit beside a wrong or partial statement (Gao et al., 2023b; Kaplan & Lerner, 2016; Thorne et al., 2026). What remains unresolved is what survives the whole chain: whether fixed rules for identity, time, missing values and source support carry the intended meaning through, and hold back a claim they cannot support. RQ1 therefore needs labelled outcomes at each step, claim support, complete source records and agreement across repeats, not a demonstration that a report can be produced (Nikiforova et al., 2020; Pineau et al., 2021).

For RQ2, splitting the work between named roles can make responsibilities visible, but surveys and trace analysis also find failures of coordination, verification and system design; one program applying the same fixed gate remains a credible alternative (Cemri et al., 2025; Guo et al., 2024). What remains unresolved is whether a separately recorded verifier changes which claims are admitted when the inputs are equal, and at what cost. RQ2 therefore compares conditions with and without that verifier using supported-claim precision, recall, unsupported-claim rate, abstention, failures, time, tokens and cost (Peffer et al., 2007; Pineau et al., 2021). A favourable result cannot be attributed to agent labels without a comparison that isolates the rule and separation.

This choice of evaluation instrument needs its own justification, because a small set of designed cases is easily mistaken for a weak substitute for a large sample. Ribeiro and colleagues argue the opposite for behavioural testing: held-out accuracy on a single test set overstates real capability, and a structured suite of targeted cases, each built to probe one specific behaviour, exposes failures that aggregate scores conceal (Ribeiro et al., 2020). That is the warrant for the D0 design in Chapter 3. It also fixes what such a suite can conclude, and here it sits usefully against Kapoor and Narayanan’s account of over-optimistic machine-learning evaluation, which shows how easily designed or leaky setups flatter a system (Kapoor & Narayanan, 2023). Read together, the two support a specific and limited position: a behavioural suite is diagnostic rather than estimative, so passing it demonstrates that a targeted control is present and cannot establish a rate on unseen data.

For future human evaluation, guidance supports correction and decision authority, while controlled evidence also shows usability and over-reliance trade-offs (Amershi et al., 2019; Buginca et al., 2021). Commercial pages describe dashboards and reporting workflows, but not independently measured reviewer effort or decision quality (Visible, n.d.; Zapflow, n.d.). Whether named review changes correctness, completeness, editing effort or elapsed time therefore remains a pilot question. An implemented approval screen cannot answer it.

Public-web research remains a separate engineering case, because retrieved content carries both prompt-injection and weak-support risks (Gao et al., 2023a; Greshake et al., 2023), and Figure 2.2 separates finding a source from accepting it as evidence. The live comparison between official sources alone and LLM-assisted search was never fixed before evaluation, so it is treated as neither a question nor a result (OpenAI, n.d.-b; Pineau et al., 2021). Chapter 3 turns these gaps into the fixed evaluation method.

Approach	What it can provide	Main trade-off for this prototype	Evidence position
Manual workflow	Flexible human collection, reconciliation and judgement	Provenance, timing and review depend on the local procedure	No measured baseline in this study
Deterministic pipeline	Repeatable rules, typed data and inspectable lineage	Limited handling of new unstructured evidence; wrong rules repeat consistently	Used for structured processing, not evaluated as a complete alternative
Single grounded agent	One interaction path can retrieve, extract and apply a fixed support gate	Production and checking share context and permissions unless separated explicitly	Credible alternative; not compared empirically
Role-separated workflow	Narrow permissions, persisted hand-offs, failure isolation and an approval boundary	More states, messages, storage and maintenance	Selected for inspectability; D0 isolates the verification gate, not general multi-agent superiority
Managed agent platform	Configured persistent roles, tools, context and hand-offs without building every interaction surface	Provider-managed boundaries and shared computer state require separate control review; setup and maintenance effects remain unknown	Grok Bot is a future comparator; not implemented or evaluated

Table 2.2. Architecture alternatives for the reporting problem

Source note: Author synthesis from Buneman et al. (2001), Nikiforova et al. (2020), Gao et al. (2023c), Guo et al. (2024), Wu et al. (2024), Cemri et al. (2025), Dibia et al. (2024) and SpaceXAI (2026). The table compares design choices; it does not report an effectiveness comparison.

Research Design and Methodology

3.1 Applied design-science strategy

This dissertation uses an applied design-science strategy because the research object is a working portfolio-reporting artefact for a bounded organisational problem. Design science links construction to evaluation against stated objectives (Hevner et al., 2004; Peffers et al., 2007). The method therefore separates three kinds of evidence: implementation records show what was built, engineering tests show which specified behaviours were exercised, and D0 shows how two workflow conditions handled labelled fictional cases. None of these observations is treated as evidence of business benefit.

The student built and evaluated the artefact, which creates confirmation risk. The main controls were fixed case labels and metrics, versioned code and fixtures, saved result files, complete failure accounting and repeated runs. Reproducibility requires enough code, data, procedure and reporting detail for another person to inspect the route to a result (National Institute of Standards and Technology, 2023; Pineau et al., 2021). These controls make changes visible, but they do not provide independent validation.

Figure 3.1 shows the evidence boundary used throughout the study. Manual work, participant review, live-web operation and managed-platform comparison were removed from the active research questions because they had no authorised observations. Their protocols are retained only as future evaluation material. This keeps unobserved outcomes null rather than converting an evaluation plan into a result (Hevner et al., 2004; National Institute of Standards and Technology, 2023).

3.2 Evaluand and comparison conditions

The evaluand is the complete local workflow configuration: input rules, catalogue, source access, operational roles, verification, prompts, schemas, computing environment and stop conditions. This binds each observation to one defined artefact version. Design-science evaluation requires comparison with stated objectives (Hevner et al., 2004; Peffers et al., 2007), while system reporting should state the intended use, version and evaluation procedure (Mitchell et al., 2019).

The executed comparison uses C1 and C2. C1 applies the shared catalogue and normalisation rules and emits each eligible cleaned value unless identity is unclear. It retains available provenance but has no separately recorded verifier. C2 receives the same candidate values and evidence, then admits a claim only when the separate fixed verifier marks it as supported. Composition cannot override that decision. Each condition ran three times on the same fourteen D0 cases, so the measured difference is

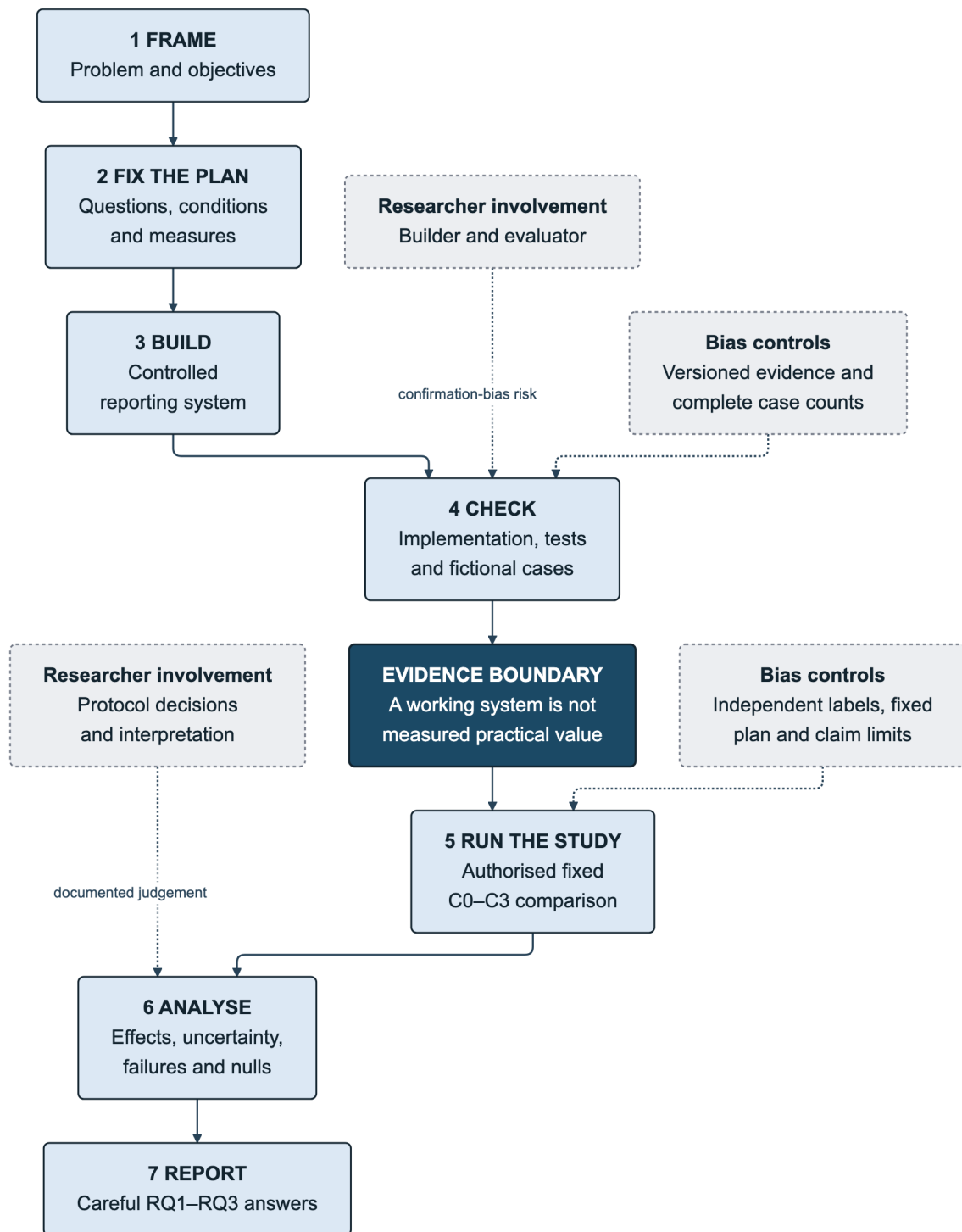


Figure 3.1. Boundary between construction, engineering checks and study results. Developed by the author from the cited methods literature and the project's fixed rules.

the verification gate and its recorded decision, not a claim about agent personalities (Guo et al., 2024; Wu et al., 2024).

The original protocol also defined C0 manual work and C3 named review. These conditions did not run and are not used to answer RQ1 or RQ2. Their detailed condition matrix is kept as Table D.1 in Appendix D to support a future authorised pilot. Retaining the protocol preserves the planned comparison without presenting missing observations as evidence (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

3.3 D0 fixture, labels and freeze

D0 contains fourteen visible, labelled fictional cases designed to expose known error categories. They cover complete and sparse records, blank and observed-zero values, mixed types, conflicting and stale evidence, unsupported and inaccessible sources, unclear identity, duplicate support, narrative evidence, mixed currencies and prompt injection. This one-case-per-category structure is deliberate and follows the behavioural-testing argument set out in Section 2.7: a suite of targeted cases, each probing one specified behaviour, reveals failures that an aggregate score over a single test set hides (Ribeiro et al., 2020). Synthetic material serves that purpose well because it supplies controlled labels, provided its construction and permitted use are explicit (Bradley et al., 2024; Gebru et al., 2021).

The fixture and its evaluation record are checksum-pinned. Loading checks the schema version, data classification, source version, file checksum and test-only namespace. Case and evidence identifiers are placed in a benchmark namespace that the operational collector rejects. The labels specify the expected emission decision and verification status. The complete case ledger is Table F.2 in Appendix F (Gebru et al., 2021; Pineau et al., 2021).

D0 is a development fixture, not a sample of companies. The rules and cases were developed together, so the result establishes only whether the programmed gate behaves as designed on these cases. A future pilot must create independent labels before a sealed final set is opened. Separating development from final evaluation reduces leakage risk (Kapoor & Narayanan, 2023), and versioned data records support later repetition (Mitchell et al., 2019; Pineau et al., 2021).

3.4 Source parity and reference decisions

For D0, both conditions receive the same candidate value, evidence record, reporting period and precondition. Scoring uses the fixture’s expected emission and status fields. The evaluator does not give either condition later evidence or a wider source window. This equal-input design is sufficient for the narrow C1/C2 mechanism comparison, but the visible development labels are not an independent reference (Kapoor & Narayanan, 2023; Pineau et al., 2021).

A real-company evaluation would require an eligible-source list, cut-off date, independently labelled claims and recorded adjudication. Agreement would indicate consistency rather than truth because reviewers can share an error (Artstein & Poesio, 2008). Exact locators and source records would also remain necessary because citation presence does not establish complete or correct support (Gao et al., 2023a; Pineau et al., 2021). The full prospective reference-parity matrix is Table D.2 in Appendix D.

3.5 Measures and denominator rules

The primary D0 measures are supported-claim precision, recall, F1, unsupported-claim rate, supported-claim rate, source-record completeness, verification or abstention accuracy, contradiction accuracy and exact repeat consistency. Precision uses supported emitted claims over all emitted claims. Recall uses correctly emitted required claims over all required claims. Unsupported claim rate uses unsupported emitted claims over all emitted claims. These denominators keep claim correctness separate from citation presence and support (Gao et al., 2023a;b).

Schema and normalisation accuracy are also reported because both conditions share earlier processing. Contradiction accuracy has two applicable cases and is interpreted as a small case result, not a stable rate. Duration values below one millisecond were truncated to zero, so they cannot support a speed comparison. Model cost is zero because D0 uses fixed local rules. Measures without D0 labels remain null, not zero. Clear missing states and denominators are necessary for task-specific data-quality interpretation (Geburu et al., 2021; Nikiforova et al., 2020).

The complete metric and denominator matrix in Appendix F also defines prospective human time, edit, usability, source-coverage and cost measures. Those fields are pilot metrics, not results of this study (Gale et al., 2013; National Institute of Standards and Technology, 2023).

3.6 Analysis and interpretation

The evaluator reports counts before rates and keeps all fourteen paired cases in both conditions. The first repeat supplies the scored values, and the complete ordered result list is compared across all three repeats. Individual cases explain why an aggregate changed. No confidence interval or hypothesis test is reported because D0 is a small, deliberately constructed fixture rather than a sample drawn from a target population (Demšar, 2006; DiCiccio & Efron, 1996).

Error analysis uses the expected verification states and records false emission, false withholding, conflict, stale evidence, inaccessible evidence, identity hold and untrusted content separately. Negative and unavailable outcomes remain in the record. Any later study should pair the same company-periods, report exclusions before rates and use uncertainty at the company or report level, not treat repeated fields as independent observations (Kapoor & Narayanan, 2023; Pineau et al., 2021).

3.7 Ethics, leakage prevention and reproducibility

Every input is classified as restricted, internal, public or synthetic. Restricted material remains local, and public availability does not itself authorise every use. Participant observation and live external processing require separate authority, purpose, consent and storage decisions (Central Digital and Data Office, 2023; Geburu et al., 2021; National Institute of Standards and Technology, 2023). D0 uses only fictional data and no external source or model call.

The evaluation record retains the Git revision and dirty-state warning, platform and dependency versions, migration version, fixture and protocol checksums, test commands, result files, failures and output hashes. A checksum detects byte changes but cannot establish truth or independent replication (Mitchell et al., 2019; Pineau et al., 2021). The detailed implementation snapshot is in Appendix G.

Collection or evaluation must stop when authority is unclear, a credential is exposed, unexpected

personal data appears, source terms would be breached, restricted data would reach an unapproved service, or a sealed evaluation set is compromised. The detailed evidence-control matrix is retained in Appendix E. These controls define the research boundary; they do not imply ethics approval for the future pilot (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

System Design and Implementation

4.1 Requirements and trust boundaries

This chapter describes what was actually built, working outward from the trust boundary to the controls that sit on each stage. The release is a local, single-user research prototype rather than a service. It listens only on the machine that runs it and refuses any client or host header that is not local (NFR-SEC-001, FR-UI-003). The bounded company-research route accepts a Companies House number on its own, with no company name or website (FR-INT-001), but a named person must review and accept that number before research can begin (FR-RES-001). One reviewed number is therefore the whole of what this route needs to start. It fixes which registered company the work refers to, which is a narrower thing than knowing that the company trades (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023; National Institute of Standards and Technology, 2023).

Every item entering the system is labelled public, internal, restricted or synthetic, and each case keeps the purpose and the person behind it. The charter assigns the roles: the student operates the prototype, portfolio companies keep their confidentiality interests, and data owners or the university ethics body authorise any restricted or participant use. External-model processing is off by default and, when switched on, accepts only public or synthetic evidence (FR-EXT-005). The client sends `store=False` with every request, which controls what the provider saves as application state but not what it retains for abuse monitoring; Section 7.2 records the requests that were actually made under this route (Central Digital and Data Office, 2023; Gebru et al., 2021; OpenAI, n.d.-a).

No output leaves the system without a person deciding that it may. Export of a report is blocked until the current version is approved (FR-REP-005), and reaching a company deck requires both a named review and a content-hash check (FR-RES-007). Four things sit outside the design entirely: remote or multi-user deployment, profiling of individuals, speculative valuation, and buy or sell recommendations. Direct live Companies House retrieval remains held at gate G2. The public-web route is built, and the external-model extraction provider was exercised in the bounded development smoke runs recorded in Section 7.2, but live public-web discovery, live capture and the frozen evaluation have not run. Table C.1, in Appendix C, maps each control to the risk it addresses, chiefly misleading communication and untrusted content (Financial Conduct Authority, 2024; Greshake et al., 2023; National Institute of Standards and Technology, 2023).

4.2 Architecture and local deployment

The artefact runs as two local processes. Next.js renders the operator dashboard, and its server-side proxy calls a private FastAPI service, so the browser never reaches the API directly. Behind the API handlers, a single runtime object assembles the importing, source, workflow, reporting, intake and company-research services, and the command-line interface builds that same runtime, so both entry points exercise identical domain code rather than parallel implementations (Hevner et al., 2004; Peffers et al., 2007).

SQLite holds the canonical identity, workflow, claim, review and audit records, with foreign keys enabled. Startup migrates the database to the declared head rather than creating tables directly, so the schema always has a stated version. Imported bytes and source captures are written once into local directories that Git ignores, verified by checksum and recorded by path, and approved exports are staged as manifest-backed bundles. Keeping relational metadata apart from the evidence bytes means a record can be traced to the exact file it came from (Gebru et al., 2021; Pineau et al., 2021).

Docker Compose mirrors that two-process shape in pinned images. Only the dashboard is published to the local machine; it reaches the API over a private container network, both container file systems are read-only, and the API alone mounts the volume holding persistent state. Health, proxy, host, cross-site request forgery and migration checks exercise these mechanisms, and Figure 4.1 maps the resulting request and data path (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Every control described here is a local one, sized to the attack surface of a single laptop rather than a hosted service; Section 7.5 states what a remote or multi-user release would additionally require (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

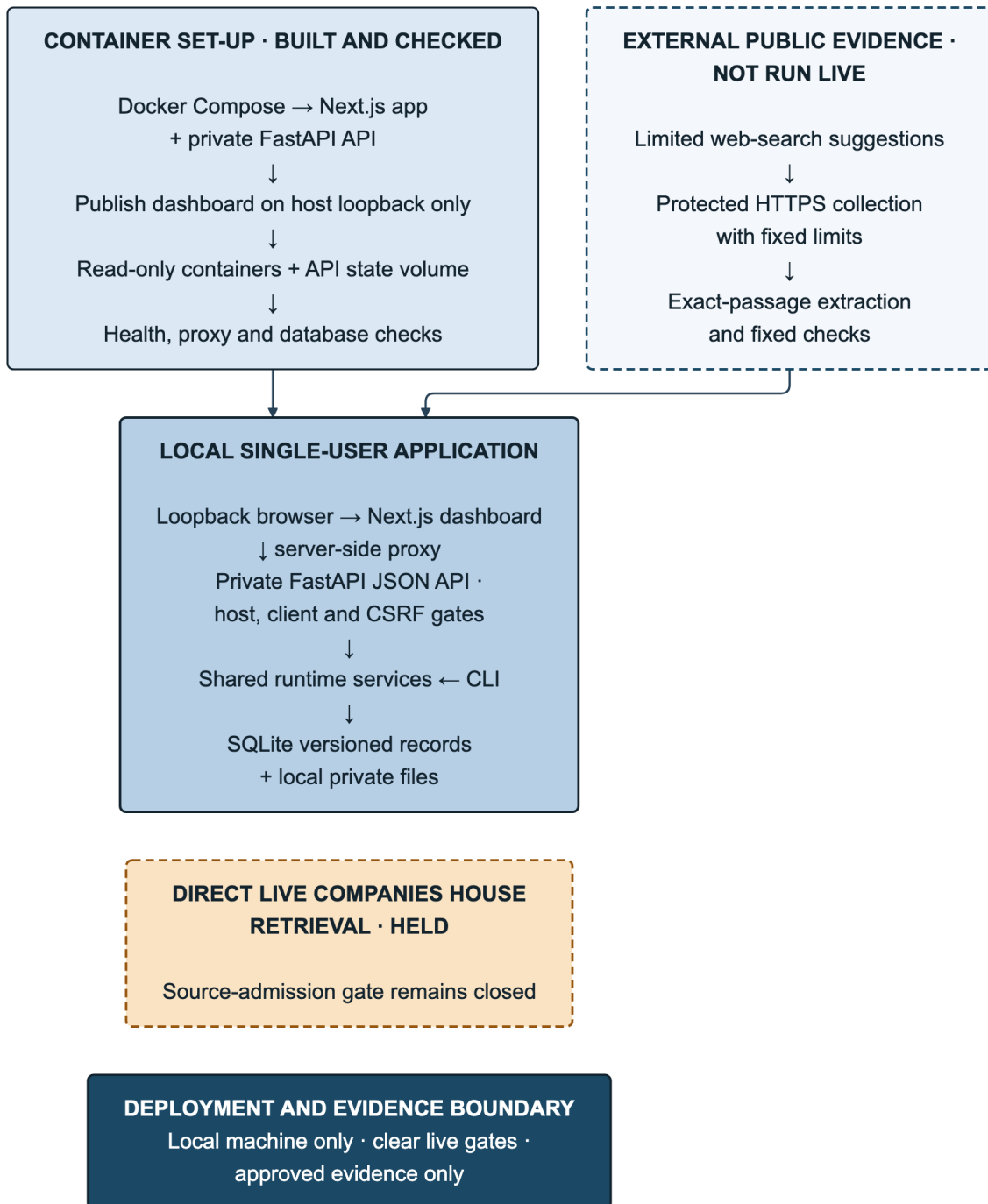


Figure 4.1. Local architecture, deployment and external-boundary status. Author synthesis of the repository’s contracts, its implementation and the local verification evidence.

4.3 Shared data definitions, storage and source records

Shared meaning starts from a versioned catalogue. Each metric definition fixes a key, type, unit, sourceability, permitted aliases and one period meaning, and a hash over the catalogue stops a version being reused after its meaning has changed. Observations then bind a company, metric, period and raw submission together, and normalisation keeps the original value beside the cleaned one. Blank, explicit zero, none stated, not applicable, not reported, not found publicly, source unavailable and invalid are eight separate states, not interchangeable nulls. Distinguishing them is what allows a later reader to see why a figure is absent, though the eight categories are this project’s definitions and would need a domain owner’s agreement before wider use (Gebru et al., 2021; Nikiforova et al., 2020).

SQLite separates canonical companies and identifiers from submissions, observations, reporting periods, evidence, claims and reports. Foreign keys, uniqueness constraints and indexes protect relationships, while Alembic revisions 0001–0010 reproduce the ORM schema and incrementally add provenance, company-research and reviewed group-scope records. Revision 0010 was added after the dated evaluation snapshot, which Chapter 5 records at 0009; the results in that chapter therefore describe the earlier schema. Tests compare the migrated schema against the one the code expects, and exercise downgrade preflights. One weakness is visible here: many of the semantic states above are stored as strings or JSON and are enforced by application code rather than by the database, so the schema alone cannot guarantee they are used consistently (Gebru et al., 2021; Pineau et al., 2021).

Imported files are hashed before persistence and published create-once under checksum-addressed paths. Connector snapshots record source and version, reviewed identifier, cutoff, locator, media type, checksum, retrieval and publication times, and classification. Facts retain value, unit, currency, periods, effective and publication times, exact structured locator and extraction method/schema. A versioned derivation hash covers terminal snapshot metadata, facts and events, detecting same-byte parser drift. The public-web case implements redacted capture and checksum verification, though no live capture has run. Writing files once and hashing them means a later reader can tell whether the evidence behind a claim has changed since it was collected (Gebru et al., 2021; Pineau et al., 2021).

Claims persist normalised values and link many-to-many with evidence items; each evidence item carries publisher, locator, times, checksum, connector version, classification and trust or time flags. A separate verification stage records supported, contradicted, insufficient, stale or untrusted outcomes without rewriting the source. Portfolio reports can export these links without raw evidence. The separate company-research schema stores accepted passages word for word alongside their source URLs, and where two sources disagree it groups the statements as contradiction candidates for a named person to resolve. No live company has been run through that path (Gao et al., 2023a;b).

Reports sit at the end of this chain and are versioned by appending rather than overwriting, so an edit retires the current section instead of erasing it and recalculates the report’s content hash. Approval is therefore bound to an exact version of exact content: change the content and the approval falls away. Section 4.8 describes the review and export controls that enforce this, and Figure 4.2 maps the boundaries (Amershi et al., 2019; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

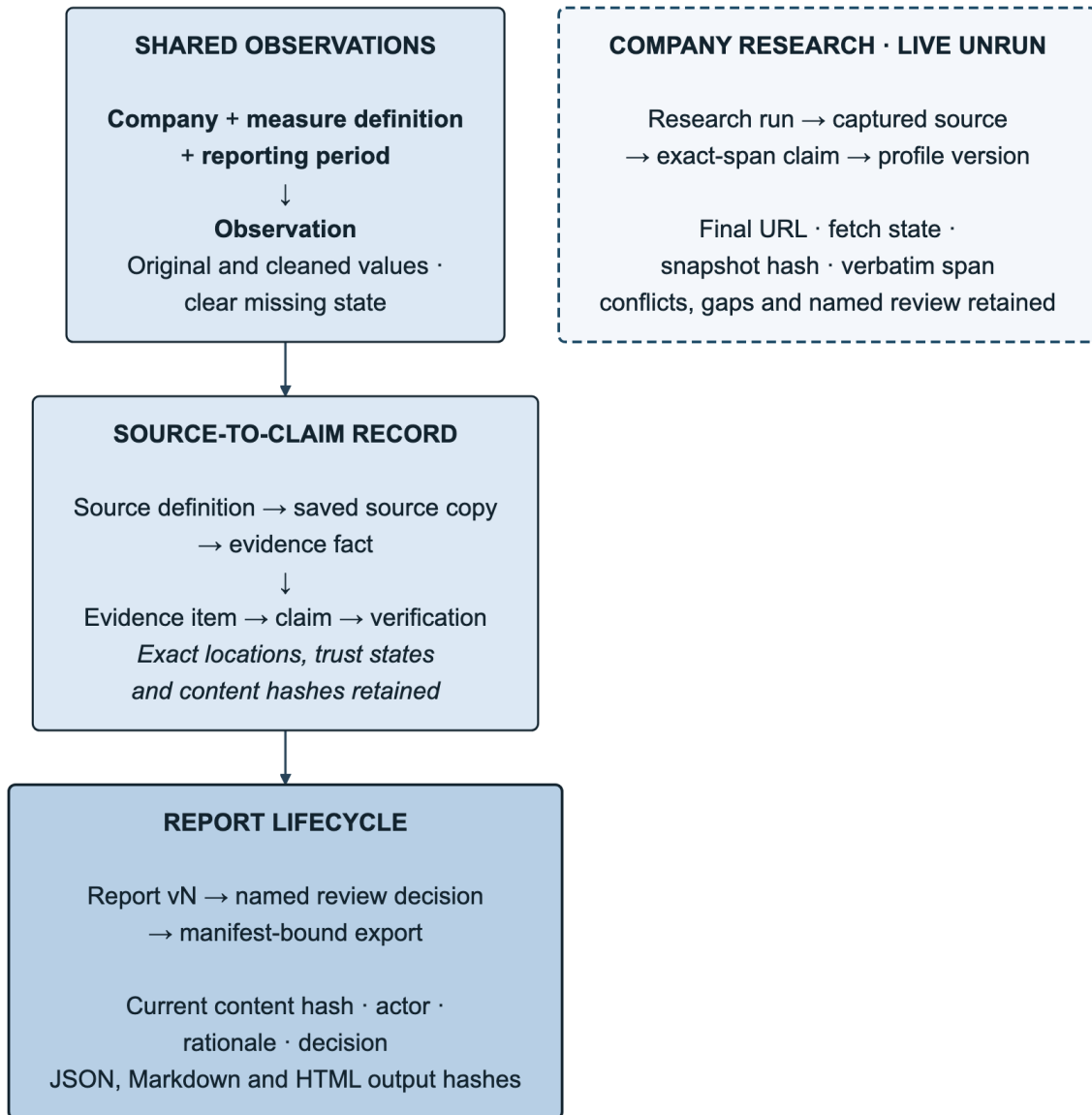


Figure 4.2. Canonical data, provenance and approval lineage. Author synthesis of the implemented relational contracts. The dashed panel is the company-research track, which runs on its own records rather than as a stage of the portfolio chain.

4.4 Intake and legal-identity control

The portfolio-import route accepts XLSX, CSV and JSON, then hashes the original bytes before writing a create-once snapshot. Its dataset identifier combines those bytes with the reporting-period label; a repeated hash and period reuses the stored submission only when classification, cutoff and profile remain compatible. Every company row then passes through the same identity resolver, and a conflicting identifier becomes a counted hold rather than a silent repair or merge (Gebru et al., 2021; Pineau et al., 2021).

The company-intelligence route accepts a Companies House number without a name or website, tidying its format before checking its structure. A well-formed number creates an exact identifier, a placeholder company and a held case; crucially, it is not confirmed against the register, because valid shape is not proof of existence. Website, name-and-jurisdiction and document routes instead store what was submitted as a claim, leaving domains pending and document bytes untrusted and checksum-bound, since legal registration and operational identity are not the same thing (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Identity becomes ready only after a named actor accepts the exact identifier with a substantive reason before source use. A rejection leaves identity unresolved; an identifier/name disagreement, a reused name with another number, a cross-classification collision, or a domain verified elsewhere is refused or held. Names may suggest a candidate, but neither name normalisation nor fuzzy similarity merges companies; name-only portfolio imports create unresolved records for review. This conservative choice sacrifices recall to reduce false joins, consistent with evidence that heterogeneous registers need cautious linkage and retain matching error (Galanakis & Savagar, 2026; Surak & Inkley, 2026; Thorne et al., 2026).

A fingerprint over intake content, purpose, classification and contract versions means an exact repeat reuses the stored artefact rather than creating a duplicate, and source collection will not begin unless the reviewed number matches both the source key and the company. Figure 4.3 maps these gates, and direct live Companies House access remains held at gate G2 (Galanakis & Savagar, 2026; Pineau et al., 2021).

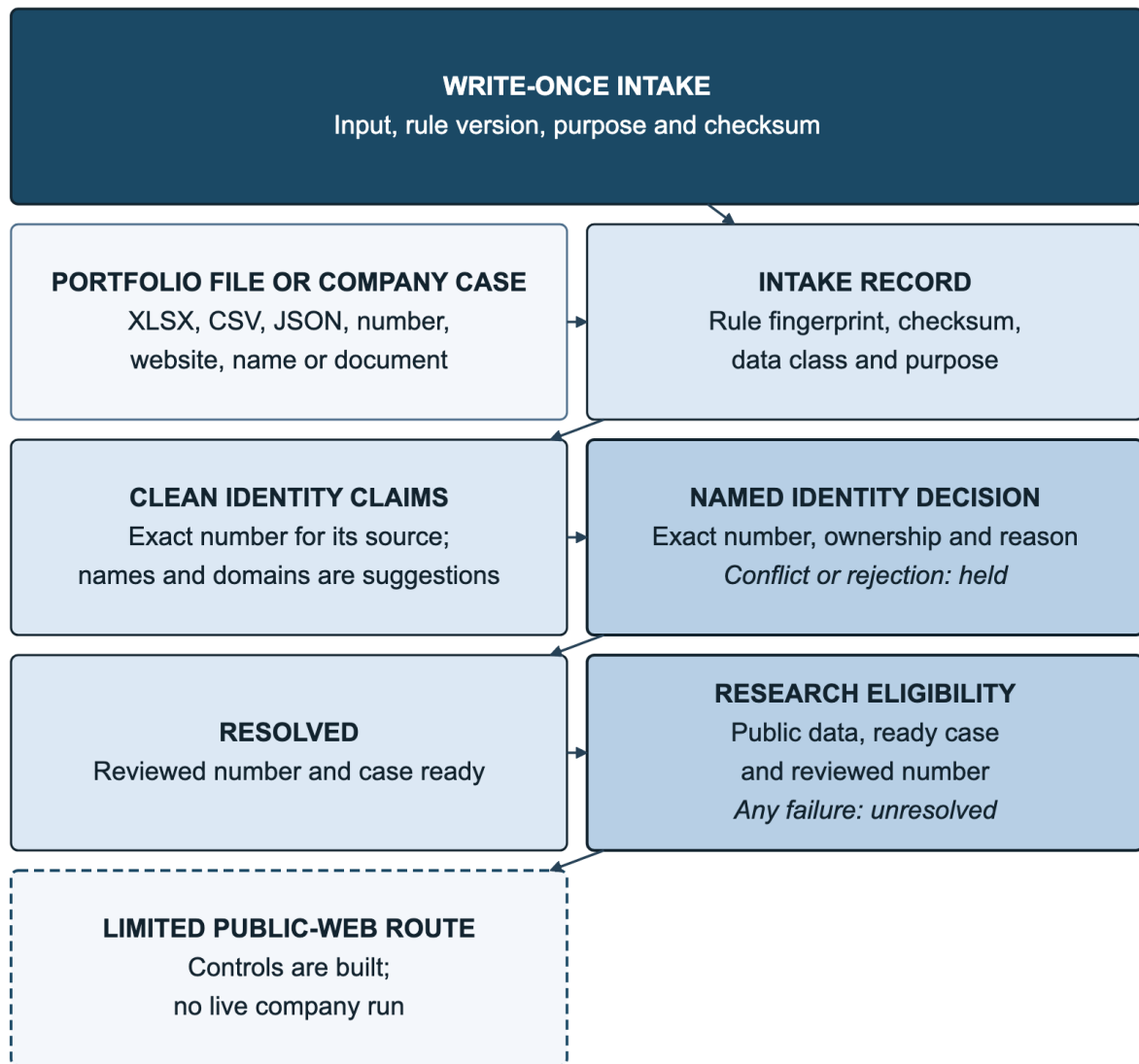


Figure 4.3. Legal-identity intake, decision and downstream research gate. Author synthesis of the implemented local controls.

4.5 Fixed portfolio workflow and separate verification

The implemented portfolio workflow follows eight fixed stages: plan, resolve identity, collect evidence, extract values, normalise them, verify claims, compose a report and stop at human review. Each stage has a named functional role and a persisted run record holding input and output hashes, status, timing and attempts. A failure is recorded and stops the stages that would have followed it, and unresolved identity stops collection before it starts. Because every stage leaves a record, a reader can later ask not only what the workflow produced but where it stopped and why (Hevner et al., 2004; Pineau et al., 2021).

Collection stores run-linked evidence and its time decision before extraction. Only temporally eligible, trusted public evidence reaches the extraction provider; untrusted material is counted and bypassed. A strict schema binds extracted company, metric, period, value and exact evidence locator, while deterministic metric rules then normalise type, unit, currency and missing state. Existing extractions are reused and every provider attempt is stored, including the failed ones (Gao et al., 2023a;b; Pineau et al., 2021).

Verification is the stage that decides what may be said. Before a claim is constructed, the workflow checks sourceability, provenance, period, trust, value and currency, then writes a verification record beside the claim. The verifier is a pure function returning one of five states: supported, contradicted, insufficient evidence, stale, or rejected as untrusted. A conflict between two current public sources cannot be averaged away into a compromise figure. Crucially, the verifier runs separately from the extractor and the composer and cannot rewrite their records, so a claim can be refused but not quietly repaired. This is separation of duties inside one program, not independent judgement by a second party (Gao et al., 2023a;b; National Institute of Standards and Technology, 2023).

Composition places supported claims in the narrative and exposes other states as exceptions, then hashes the current, versioned report and moves it to pending review. A named reviewer may approve, reject or request an edit; each decision records actor, reason and report version. Editing creates a new section version, recalculates the content hash and revokes approval. Export remains unavailable until current approval, and optimistic locking plus a manifest-backed staged write guards stale or partial publication attempts (Amershi et al., 2019; Bućinca et al., 2021; Pineau et al., 2021).

Two design choices are worth separating here, because they are often confused. The first is fixed orchestration over open-ended agent conversation: the workflow is a state machine with explicit stops, which removes uncontrolled branching and puts every failure in a locatable place, at the price of any adaptive planning. The second is role separation, which Section 1.5 argues for and this chapter implements. Its value lies in who may do what, not in how many components exist: the extractor cannot admit its own output, the verifier cannot rewrite what it judges, and neither can approve a report. Consistent with that section, a single program could apply the identical support rule, and the D0 comparison in Chapter 5 isolates that rule rather than the number of roles. Figure 4.4 maps these boundaries (Guo et al., 2024; Peffers et al., 2007; Wu et al., 2024).

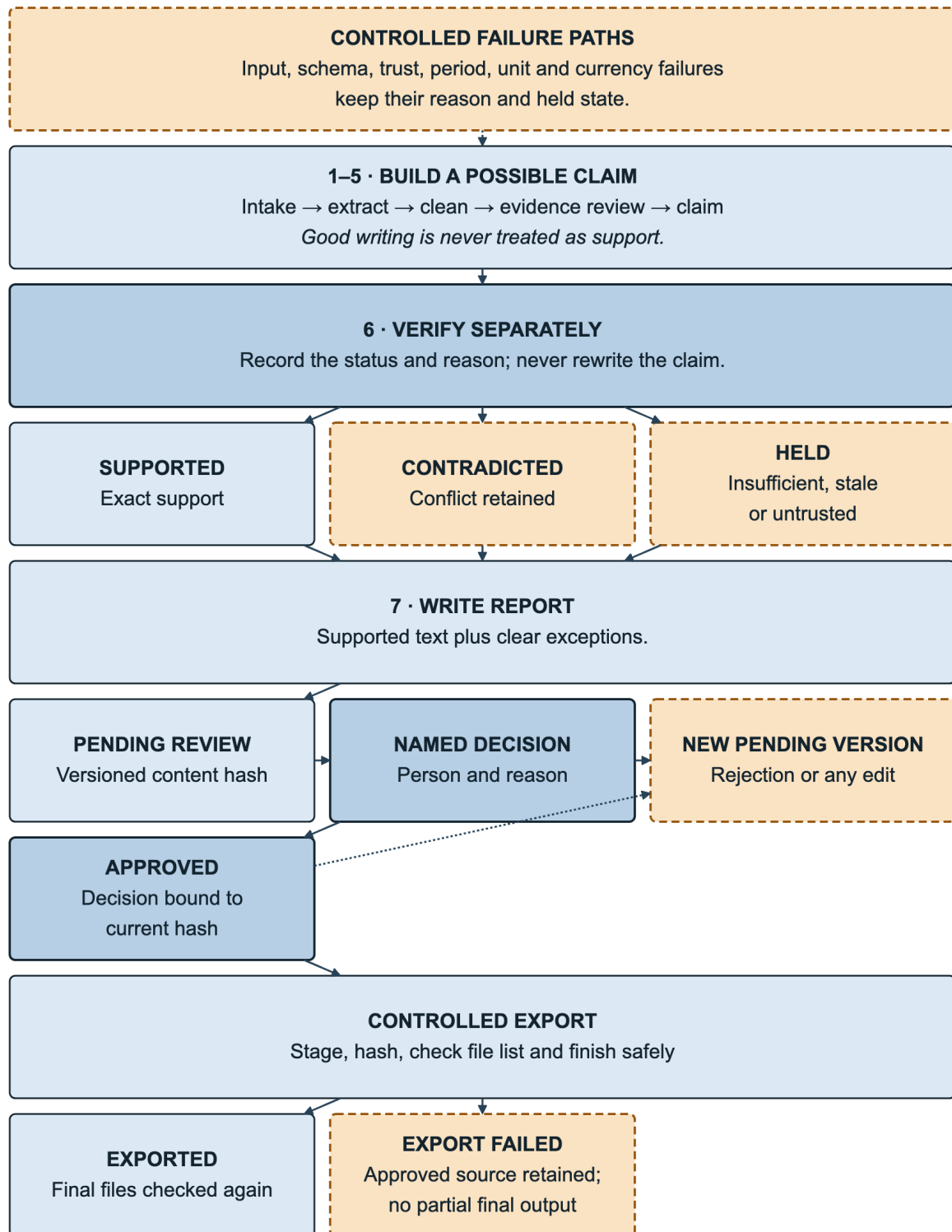


Figure 4.4. Fixed workflow, separate verification states, named approval and controlled export.
Author synthesis of the implemented local controls.

4.6 Fixed source connectors, time and quality

The source layer is a registry of declared capabilities rather than a crawler. Every request names the source, company, reviewed identifier, cut-off, fact keys and mode, and the registry refuses it before any adapter runs if the capability is unsupported, the identity is unresolved, unreviewed or expired, or live retrieval is requested while admission is closed. A refusal of this kind is a decision not to look, which the system records differently from a publisher having no record (Gebru et al., 2021; Pineau et al., 2021).

Two deterministic adapters are usable, both through synthetic offline replay. The Companies House adapter (manifest 1.4.0) matches an exact number and exposes company, filing and charge fields through JSON locators. The UKRI adapter (manifest 1.3.0) matches an exact organisation identifier, selects the most recent correction available at the cut-off, and emits stable project and outcome locators. Each replay stores the source version, the bytes, a checksum, the retrieval and publication times, and a derivation hash, so the same input can be re-examined later and shown to be the same input (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023; Pineau et al., 2021).

Time is judged relative to the claim, not to the moment of collection. Evidence is ineligible if it carries no recorded publication or availability time, was published after the reporting cut-off, takes effect in the future, or had expired before the period began; when it was retrieved is tracked separately. A cumulative figure must span exactly from the stored programme start to the cut-off. UKRI funding binds only when every award in the window carries a finite, explicit sterling amount, and otherwise leaves a missing state with a diagnostic rather than an inferred zero or an unauthorised currency conversion (Hardman & Ramírez Santos, 2023; Kapoor & Narayanan, 2023; Nikiforova et al., 2020).

The reasons a fact can be absent are then kept apart, because collapsing them would destroy the very distinction the report depends on. A lookup that finds nothing is `no_record`; a source that could not be reached is `source_unavailable`; a request refused before the adapter ran is a policy block; a value that fails validation is a typed missing state; and a collection problem that cannot be retried is `failed`. The quality contract built on these states excludes untrusted or time-ineligible evidence, holds incomplete provenance and conflicts, and warns on the bounded absence cases. Table C.2, in Appendix C, maps the decisions, all of which remain bounded by the snapshots admitted, with live Companies House and UKRI retrieval held at gate G2 (Gao et al., 2023a; Gebru et al., 2021; Nikiforova et al., 2020).

4.7 Bounded public-web company-research case study

This section describes the one route that reaches outside the local machine, and the point of its design is that search is allowed to suggest but never to supply. The path opens only after a public case has a reviewed Companies House number and a fixed reporting cut-off. The Responses API then runs `web_search` with `store=False` to plan discovery, and the application admits only canonical HTTPS addresses that also appear in the tool’s own list of consulted sources. Page titles and model prose stay metadata; they never become company facts (Gao et al., 2023a; OpenAI, n.d.-b).

Application code, not the search result, then fetches each candidate. It refuses embedded credentials, non-default ports, local hosts and addresses that do not resolve publicly; it checks robots rules, re-validates every redirect, permits six textual content types, and enforces limits on elapsed time, redirects and bytes. The connection is made to a verified public address while still checking the hostname on

the certificate. Every page that comes back is treated as hostile input, because hidden instructions can travel inside remotely supplied content (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

What survives capture is converted to visible text, stripped of personal contact details, and stored under a content hash in a directory that refuses to overwrite. Extraction sees only that bounded, redacted text, and a proposed statement is admitted only if its address matches a captured source, the statement equals the evidence passage, and that passage occurs literally in the stored text, with the cut-off rule satisfied and any date or amount appearing inside the passage itself. A citation therefore becomes something a reader can check rather than a mark implying that checking has happened (Gao et al., 2023a;b; National Institute of Standards and Technology, 2024).

Four tasks run in order behind this – discover, capture, extract and compose – under one fingerprint covering identity, cut-off, model, prompt, policy and budgets, so no task can start before its prerequisite or resume under a changed contract, and every retry, token and byte is counted against the run. Composition is then fixed rather than generative: it groups admitted claims, marks multi-source disagreements for a named person, and lists the categories that were blocked, unsupported, failed or simply not covered, so a gap stays visible instead of being smoothed out of a persuasive narrative. Figure 4.5 summarises the funnel. Everything described here ran against fake providers and controlled responses, which makes this route implementation evidence that answers neither research question (Gao et al., 2023a; National Institute of Standards and Technology, 2023; 2024).

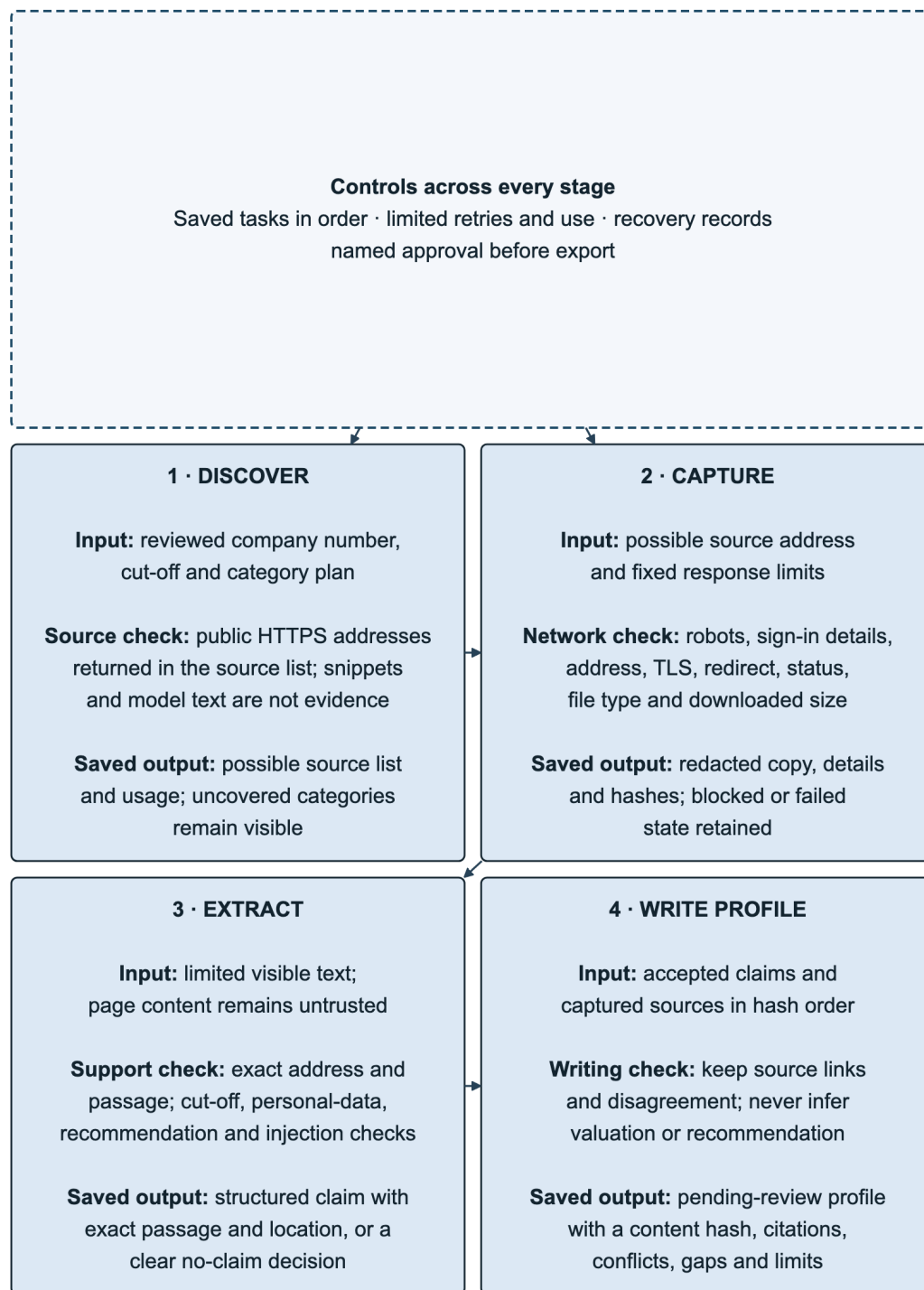


Figure 4.5. Limited public-web evidence process, built and tested on fictional inputs but never run against a live company. Mermaid diagram developed by the author from the implemented controls.

Source note: OpenAI (n.d.-b, PDF pp. 1–3, 6) documents Responses web search and source URLs; Gao et al. (2023a, PDF pp. 1–3, 8–9) shows that citation quality and factual correctness require separate assessment. Greshake et al. (2023, PDF pp. 1–5, 25) and NIST (2024, PDF pp. 15–16, 44, 48) support treating downloaded content as untrusted and keeping risk controls, source records and human review. These sources do not certify this implementation or establish live performance.

4.8 Human review and controlled outputs

The portfolio-report path stops at pending review. Approval or rejection requires the configured reviewer identity, a substantive rationale and the current optimistic-lock token. Nobody can approve a report until every claim in it carries a recorded verification, and any edit appends a new section version, recalculates the report hash and cancels the previous decision. The effect is a deliberate stopping point at which a named person, not the software, takes responsibility for release (Amershi et al., 2019; Bućinca et al., 2021).

Company-research profiles use a separate, narrower gate. Before a named approve or reject decision, the service rechecks the profile hash, pinned run contract and completed composition task; stale locks, altered content and unfinished recovery are refused. Contradiction candidates and coverage gaps stay inside the dossier the reviewer sees, so a gap has to be looked at rather than discovered later. Only approved profiles can be downloaded, as JSON or printable HTML (Gao et al., 2023a; National Institute of Standards and Technology, 2023).

For portfolio reports, export accepts only the current approved version and produces JSON, Markdown and accessible HTML. Files are staged below the local export root, bound to a versioned manifest and content hash, then re-verified before reuse, and a partial write is never marked as exported. Release is therefore local and deliberate: the system writes files to a folder, it does not publish (Geburu et al., 2021; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

4.9 Failure, recovery, and implementation-status ledger

One principle runs through every control above: a failure keeps its own name and stays visible, so that none of the distinct outcomes catalogued in Section 4.6 can quietly become a successful claim. Retryable tasks keep their attempts and final reasons, and interrupted work resumes only under the same run contract, so a retry is safe rather than hopeful (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Table C.3, in Appendix C, then separates tested engineering work from work that is on hold, and the distinction matters because a reader can otherwise mistake a working feature for a tested outcome. The local workflow, source records, approval and export controls, offline connectors and gated public-web route were built and tested with fictional or controlled inputs. Direct live official-source access, a live public-web check, the fixed company comparison, authorised human evaluation and production deployment were not run, exist only as plans, or fall outside scope (Hevner et al., 2004; Peffers et al., 2007).

Evaluation and Results

5.1 Exact implementation and evidence snapshot

The engineering record was created on 28 August 2026 using macOS Darwin 25.6.0 on Arm64 and Python 3.12.13. Git HEAD was 6356aa49cb3b, but the working tree contained dissertation and implementation changes, so the commit is not a complete identifier. Table G.1, in Appendix G, records checksums for the dependency lock, protocol, test data and migrations, and its text alternative retains the full values, so another person can confirm they are working from the same inputs (Gebru et al., 2021; Pineau et al., 2021).

The three-repeat D0 output was regenerated and frozen on 1 September 2026 at `audit/d0_evaluation_2026-09-01.json`. Its SHA-256 checksum is `dcc2fbffab100dfcf034784002d63b610cd9f97a8d6e6f9218b6e7f900de9dcd`. The frozen output retains the evaluation-manifest and fixture checksums, the test-only namespace, all case decisions and every reported summary value. This resolves the earlier provenance gap without converting the designed cases into independent or real-company evidence (Mitchell et al., 2019; Pineau et al., 2021).

Evidence comprises the local system at migration 0009, 286 automated tests and fourteen fictional D0 cases. Later interface and schema revisions described in Chapter 4 were not part of this dated snapshot and add no empirical result. D1 is planned, D2 sealed, and C0/C3 have no observations. The company-research route passed controlled tests but has never been run against a live company, a live public website or an external model. The only live external-model requests anywhere in the project were bounded development smoke runs on the portfolio extraction stage, described in Section 7.2, and they contribute no result here. What follows is therefore an engineering record and a set of designed mechanism cases. Sections 5.2 and 5.5 supply the evidence for RQ1, which asks how the workflow preserves identity, time, missing states, provenance and approval boundaries from input to report; Section 5.4 addresses RQ2, which asks what a separately recorded verification stage does to claim admission. Conditions without observations are reported as null rather than estimated, and Section 7.5 states the boundary that applies to every number here (Hevner et al., 2004; National Institute of Standards and Technology, 2023; Peffers et al., 2007).

5.2 Engineering validation

The final engineering run used the full automated test set rather than choosing only successful cases. All 286 tests passed and executed 85.58% of the source statements. Ruff found no lint errors, and strict MyPy found no type errors across 46 source files. A new SQLite database applied migrations

0001-0009, reported 0009 as the snapshot version and needed no further Alembic changes. Together these results show that the paths the tests reach behave consistently (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

The formatting check failed. Ruff found five files that it would change, made up of two source files and three test files. The dated snapshot retained those five findings; later code changes are not evaluated by this record. The test run also produced 22 warnings about features that Starlette and the SQLite date and time adapters may remove in future. Table G.2, in Appendix G, therefore reports a mixed result instead of calling the whole run a pass. The formatting failure matters for repeatability, but it does not cancel the 286 passing tests (Gebru et al., 2021; Pineau et al., 2021).

The suite covers migrations, identity, time, source policy and records, verification, review, export, research tasks and unsafe-network controls, all using fictional data, fake providers and local files. Coverage of 85.58 per cent counts source statements that ran, which is a weaker guarantee than covering every requirement, branch or fault; browser usability, concurrent users, backup recovery and live sources were not tested at all (Hevner et al., 2004; National Institute of Standards and Technology, 2023; Peffers et al., 2007).

5.3 D0 fixture and comparison design

D0 contains fourteen labelled and fictional engineering cases, accepted through a versioned evaluation record. The cases cover complete and limited records, blank values compared with zero, mixed data types, conflicting and old evidence, and unsupported claims. They also cover unclear identity, repeated evidence, inaccessible sources, evidence in prose, mixed currencies and prompt injection. The data loader checks the file checksum, the test-only split rule and the `benchmark:d0` name. It then places case and evidence identifiers in that test namespace, so the operational collector will refuse them and no fictional company can leak into a real run (Gebru et al., 2021; Pineau et al., 2021).

C1 and C2 use the same cases, data catalogue, cleaning rules and proposed values. C1 produces every eligible cleaned value unless the company identity is unclear. It keeps the available source details but has no separate verifier. C2 uses the same preparation and produces only values that a separate, fixed verifier marks as supported. The saved evaluator output labels these conditions `deterministic_single_agent` and `multi_agent_verification`, and records the unsupported-claim rate in a field named `hallucination_rate`. This dissertation renames all three, because the one change between the conditions is the verification gate rather than the number of agents, and because a fixed pipeline that emits an unsupported value is not hallucinating in the sense that term usually carries. Each condition ran three times. The first run was scored, and the complete set of outcomes was compared across repeats. Table 5.1 makes this one change between the conditions clear. Each comparison matches the same case in C1 and C2 (Guo et al., 2024; Wu et al., 2024).

One feature of this design needs stating before any result is read. The expected outputs and verification states live in the same visible development file that informed the rules, so D0 tests programmed gates against cases written alongside them. It carries no event-link labels, no human reference and no workload measures, which is why several rows in the results are null rather than zero. A pilot on real data and a sealed final set remain the precondition for any performance claim (Kapoor & Narayanan, 2023; Peffers et al., 2007; Pineau et al., 2021).

Design element	Shared input	C1 fixed baseline	C2 verification condition
Fixture	14 fictional labelled cases; manifest and fixture hashes; test-only benchmark:d0 names- pace	Same cases and order	Same cases and order
Edge-case coverage	Complete, sparse, blank/zero, mixed type, conflict, stale, un- supported, ambiguous identity, duplicate, inaccessible, narra- tive, mixed currency, prompt injection	Same labelled inputs	Same labelled inputs
Preparation	Catalogue, whether the source can be found, candidate value and currency, normaliser and expected states	Shared	Shared
Changed mechanism	The separately recorded verifi- cation step is the only planned difference	Emits eligible standardised candidates; unclear identity is held	Emits only candidates sup- ported by the verifier; unclear identity is held
Repeats and scoring	Three fixed repeats; the first run is scored; equal result lists show repeat consistency	Three repeats	Three repeats
What cannot be con- cluded	The visible development la- bels and rules were designed together	No independent or real- company baseline estimate	No estimate of the verification effect on new cases

Table 5.1. D0 fictional cases and C1–C2 comparison design

Source note: Author summary of the runnable evaluation list, D0 cases, evaluator and fixed study plan. The table shows the implemented difference between the conditions. It does not prove that one condition is better in real use or that people would benefit.

5.4 D0 synthetic mechanism results

The repeated run produced fourteen scored cases for each automated condition. C1 produced eleven claims: five correct positive claims and six incorrect positive claims, with no required claim missing. C2 produced five claims, all correct, and returned no claim for the other nine cases as intended. C1 therefore had precision of 0.455, recall of 1.000 and F1 of 0.625. C2 recorded 1.000 for all three measures. This apparent improvement applies only to the designed D0 cases and comes from blocking the six candidates designed to be unsupported (Gao et al., 2023a;b).

The unsupported-claim rate fell from 0.545 in C1 to 0.000 in C2, while the supported-claim rate rose from 0.455 to 1.000. Complete source records rose from 0.818 to 1.000 because two C1 claims did not have every required source detail and C2 did not produce them. All three rates use the number of claims produced as their total, and that total falls from eleven to five. The result therefore shows more selective output, not wider source discovery or a more complete report (Gao et al., 2023a; Gebru et al., 2021).

Both conditions followed the schema correctly, cleaned every value correctly and produced exactly the same outcomes across the three repeats, giving 1.000 on each measure. C1 was correct in 0.571 of verification and no-claim decisions, compared with 1.000 for C2. C1 missed both labelled contradictions and C2 found both, so their contradiction scores were 0.000 and 1.000. This result is based on just two contradictions. Table 5.2 keeps that warning, and Figure 5.1 shows the main rates without combining them into one score (Nikiforova et al., 2020; Pineau et al., 2021).

Mean time was recorded as zero milliseconds for both conditions because the evaluator cuts times below one millisecond down to the nearest whole number. It cannot support a speed comparison. The recorded LLM cost was USD 0 because both routes are fixed and run locally. Measures for identity, extraction, timing of evidence, source quality, event links, report correctness and reviewer value were blank because D0 has no suitable labels or observations. Keeping a blank result instead of changing it to zero avoids suggesting a measured failure or success (Gebru et al., 2021; National Institute of Standards and Technology, 2023).

Looking at individual cases explains the overall change. C1 incorrectly produced claims for the conflict, old evidence, unsupported evidence, inaccessible source, mixed currency and prompt injection cases. C2 gave them cautious unsupported states and produced no claim, while keeping the complete case, the true zero, mixed data types, matching repeated evidence and claims supported by prose. Each category has only one case, so this shows the intended gates worked for the categories D0 contains and cannot describe a wider distribution (Gao et al., 2023b; National Institute of Standards and Technology, 2023).

Two individual cases from Table 5.3 show what these decisions look like in practice, and they are worth following in ordinary terms. In the first, the fictional company reports that it created no jobs in the period. The submitted figure is zero, the stored evidence also says zero, and the two agree, so the verifier marks the claim supported and it appears in the report with its source attached. This case matters because the obvious shortcut would be to treat a zero as an empty box and quietly drop it. Doing so would remove a real and possibly unwelcome fact from the report, so the system is designed to notice that somebody actually observed nothing and to say so (Bradley et al., 2024; Gebru et al., 2021).

The second case is a claim that was refused. A grant of one hundred thousand pounds is proposed for the period, but the stored register evidence for that same period records two hundred thousand. The two figures cannot both be right, and nothing in the evidence establishes which is. The verifier

therefore marks the claim contradicted, the report omits it, and the disagreement is kept on record for a named person to resolve. What the system specifically does not do is choose the larger figure, the newer one or an average, because any of those would manufacture a number that no source states. For a reader, the useful contrast is that C1 published this claim while C2 withheld it: the six withheld claims are the whole of the measured difference between the two conditions, and they are withheld silences rather than corrections (Gao et al., 2023a;b; Huang et al., 2023).

These results answer RQ2 for the conditions tested. The separately recorded verification stage changed claim admission: it rejected every candidate D0 designed to be unsupported, conflicting, stale, inaccessible, mixed-currency or injected, kept all five designed to be supported, and raised source-record completeness by producing fewer claims rather than better-sourced ones. The perfect scores deserve suspicion rather than celebration, because the labels and the rules were written together; Section 7.5 sets out what that shared origin does and does not allow the reader to conclude (Hevner et al., 2004; Kapoor & Narayanan, 2023; Peffers et al., 2007).

Measure	C1 baseline	C2 verifier	Total or meaning
Cases; emitted claims	14; 11	14; 5	The same fourteen paired fictional cases.
Confusion counts (TP / FP / FN / TN)	5 / 6 / 0 / 3	5 / 0 / 0 / 9	Expected-emission labels in visible D0 fixture.
Precision / recall / F1	0.455 / 1.000 / 0.625	1.000 / 1.000 / 1.000	Emitted and expected claims; evidence from designed cases only.
Unsupported / support rate	0.545 / 0.455	0.000 / 1.000	Each uses emitted claims: 11 for C1, 5 for C2.
Source-record completeness	0.818	1.000	Complete source records among emitted claims.
Schema / normalisation accuracy	1.000 / 1.000	1.000 / 1.000	Fourteen labelled cases per condition.
Verification / abstention accuracy	0.571 / 0.571	1.000 / 1.000	Fourteen labelled cases per condition.
Contradiction accuracy	0.000	1.000	Two applicable conflict cases only.
Exact repeat consistency	1.000	1.000	Three fixed result lists.
Mean duration; LLM cost	0 ms; USD 0	0 ms; USD 0	Values below one millisecond are shown as zero; both routes were local and fixed.
Other outcome layers	NULL		No applicable identity, extraction, temporal, quality, event, report or reviewer labels.

Table 5.2. D0 fictional-case results by automated condition

Source note: Three-repeat D0 evaluator output frozen on 1 September 2026, SHA-256 dcc2fbff...de9dcd (full value retained in the text alternative and Chapter 5). Values come from fictional cases. They do not support a confidence interval or a claim about performance in real use.

Case and proposed claim	Evidence	Verification rule	Decision	Report consequence
Observed zero: jobs created = 0	Portfolio submission records 0 for SYN-PERIOD; locator: submission synthetic B2	Zero is a value when the typed observation and evidence agree	Supported	Claim is emitted with its source record
Contradiction: grant funding = GBP 100	Synthetic register records GBP 200 for the same period; locator: fixture conflict	Candidate value and currency must agree with eligible evidence	Contradicted	Claim is withheld and the conflict is retained
Stale period: awards received = 1	Matching value exists only for PRIOR-PERIOD; locator: fixture stale	Evidence period must match the report period	Stale	Claim is withheld; the old evidence remains visible
Prompt injection: new partnerships = 5	Matching value appears in the fixture injection record, but it is marked untrusted	Instruction-like untrusted evidence cannot support a claim	Rejected untrusted	Claim is withheld and no instruction is followed

Table 5.3. Examples of accepted and held D0 claims

Source note: Exact fields from `fixtures/evaluation_cases.json`. All values, publishers and locators are fictional and were created for the D0 engineering evaluation.

Measure	C1: no verifier	C2: verifier
Precision	0.455	1.000
Recall	1.000	1.000
F1	0.625	1.000
Claim-support rate	0.455	1.000
Complete source records	0.818	1.000
Unsupported-claim rate ↓	0.545	0.000

Figure 5.1. D0 metric profile for C1 and C2. Author Mermaid visualisation of the current D0 output from fourteen visible synthetic cases. Emitted-claim denominators differ (C1 eleven; C2 five), so the values are neither coverage nor utility scores.

5.5 Company-research contract and adversarial results

A focused run passed 37 company-research, migration and fixture tests. The controlled end-to-end case resolved a reviewed company number, admitted bounded search suggestions and content, accepted exact supporting passages, composed profile version three and required named review before download. Other cases covered run fingerprints, replay, budgets, cancellation, recovery, source ownership, contradictions and stale approval, and each behaved as its contract specified (Gao et al., 2023a; National Institute of Standards and Technology, 2023).

Hostile-input tests rejected credential-bearing or non-HTTPS URLs, blocked ports, private or changed network addresses, excessive redirects, robots exclusions, wrong file types and oversized responses. Controlled components checked connection to a public IP with HTTPS hostname verification. Downloaded text remained untrusted; suggested URLs had to appear in the bounded source list, retries consumed the run budget, and exact captured passages supported accepted values. These selected cases do not establish complete SSRF or prompt-injection protection (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Table F.3 maps each control group to residual risk. Tests also covered publisher diversity, duplicate snapshots, identity passages, timeout failure, zero redirects and run-specific cleanup. Inputs were local fixtures, fake providers or controlled responses, with one Starlette deprecation warning. This test run contacted no OpenAI, Companies House, publisher or open website endpoint; the separate development smoke runs that did reach OpenAI are recorded in Section 7.2 and are not part of this snapshot. Consequently, delay, tokens, cost, paywalls, DNS change, source availability, open-web coverage and profile usefulness remain unmeasured (National Institute of Standards and Technology, 2023; OpenAI, n.d.-b; Pineau et al., 2021).

A caution applies to all of these security results. Each is a stated example, and a pass means one specified input was refused as designed. There is no rate here for wrongly blocked requests, no measure of attacks missed, and no coverage figure; estimating resistance beyond these examples would need generated inputs, dependency checks and penetration testing designed by someone else (Greshake et al., 2023; National Institute of Standards and Technology, 2023).

The detailed adversarial-control matrix is provided in Appendix F.

5.6 Comparisons that produced no observations

Three planned comparisons produced nothing to report: the public-company source-discovery study, the manual baseline C0 and the participant-review condition C3. They are recorded here as absent rather than as weak findings, because each needed frozen companies, reference labels and stopping rules agreed before collection, and a comparison arranged after the data are seen cannot demonstrate equal information access (Bućinca et al., 2021; Kapoor & Narayanan, 2023; Pineau et al., 2021). The held protocols and null fields are in Appendix F, and Section 6.7 sets out the pilot that would fill them. Tables F.4 and F.5 distinguish the two absent comparisons from executed results.

5.7 Negative, null, failed, and unavailable outcomes

Negative results were kept. Ruff’s formatting check failed for five files, and the passing test set produced 22 warnings about future changes. C1 produced six incorrect positive claims in D0 and missed both

labelled contradictions. Timings of zero milliseconds gave no useful information because the measure cut times below one millisecond down to a whole number. These results were not removed from the totals or presented as successful checks. They show the engineering and measurement work that remains (National Institute of Standards and Technology, 2023; Pineau et al., 2021). The formatting failure is still present in the evaluated record.

Blank outcomes were retained on the same principle. Where D0 has no label and where a condition never ran, the result field stays empty, and the distinct reasons for absence – a blocked request, an unavailable source, no record, an unsupported claim, a contradiction and an outright failure – are never merged into one (Gao et al., 2023b; Gebru et al., 2021; Hevner et al., 2004).

Discussion

6.1 RQ1: Evidence preservation in the implemented workflow

RQ1 asks how the implemented workflow preserves identity, time, missing states, provenance and approval boundaries from input to report. The repository and engineering checks show that these controls exist in one local workflow. Inputs retain dataset and period identifiers; company matches can enter an explicit hold; missing values keep typed reasons; source records retain locators and checksums; verifier decisions are persisted; and export requires approval of the current report version. This is evidence that the workflow is inspectable, not evidence that it is ready for production (Hevner et al., 2004; Peffers et al., 2007).

D0 supplies a narrower behavioural result. Both conditions normalised all fourteen fictional cases as expected and returned the same ordered outcomes in three repeats. C2 emitted the five supported cases and held the remaining nine. The accepted observed-zero case shows why a typed zero must not be collapsed into missing data. The contradicted, stale and untrusted examples show how period, value and trust rules change the report consequence. These decisions remain traceable to exact fixture records (Gao et al., 2023a;b).

Being able to trace a claim does not make its source true. A fixed workflow will faithfully repeat a publisher’s mistake, and an exact passage can support a sentence that still misleads. What the workflow offers is narrower: a reader can see where a value came from, what was done to it, why it was admitted and who released it. Judging whether the source is right, and how much of the picture it covers, are separate jobs (Gao et al., 2023b; Pineau et al., 2021).

RQ1 is therefore answered for the implemented mechanisms under controlled inputs: the workflow produces a complete record of why each claim was emitted or withheld, and Section 7.5 states the boundary of that answer (Geburu et al., 2021; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

6.2 RQ2: Observed effect of separate verification

RQ2 compares C1 and C2 on identical D0 candidates and evidence. C1 emitted eleven claims, including six designed to be unsupported. Its precision was 0.455 and its unsupported-claim rate was 0.545. C2 emitted five claims, all labelled as supported, giving precision of 1.000 and an unsupported-claim rate of 0.000. Recall remained 1.000 in both conditions. The observed effect was selective admission, not discovery of additional evidence (Gao et al., 2023a;b).

The lower claim count is important. C2 did not improve by producing a more complete report; it withheld candidates for conflicting values, a stale period, absent or inaccessible support, mixed currency and untrusted content. This was correct for D0, but a similar gate could falsely reject a valid claim in new data. D0 cannot estimate false rejection outside its visible designed cases (Kapoor & Narayanan, 2023; National Institute of Standards and Technology, 2023).

The scope of that result needs stating precisely, and it is exactly the scope Section 1.5 claimed in advance. C1 and C2 share one program, one catalogue and one normalisation route, so what differs between them is the support rule and its authority over composition, not a persona, a conversation style or a count of agents. A deterministic pipeline or a single grounded agent could apply the same rule and reach the same decisions. The finding is that an explicit, separately recorded refusal changed what the report was allowed to say; it is not evidence that multi-agent designs are generally better (Guo et al., 2024; Wu et al., 2024).

No useful speed or cost comparison was produced. Both routes used local fixed rules, model cost was zero and sub-millisecond times were truncated to zero. Role separation also adds states, hand-offs, storage, tests and maintenance. A later pilot must measure these costs alongside false acceptance, false rejection and reviewer work (Hevner et al., 2004; Pineau et al., 2021).

6.3 Why functional role separation was selected

The selected design separates production from verification and keeps approval outside both. The planner and collectors can propose work, but they cannot approve a report. The verifier can accept, hold or contradict a candidate, but it cannot rewrite the claim to make it pass. The composer can use only admitted claims, and the reviewer decides whether the current version may be exported. Persisted hand-offs make each decision recoverable and allow a failure to stop one stage without silently changing another (Guo et al., 2024; Wu et al., 2024).

This design also supports narrower permissions. Public discovery, protected collection, local restricted-data processing and final approval do not share one unrestricted tool boundary. The trade-off is more code and more operational state. Multi-agent trace studies report coordination and task-verification failures, so added roles are themselves a source of risk (Cemri et al., 2025; National Institute of Standards and Technology, 2023). The prototype uses a fixed state machine rather than an open-ended conversation to keep this cost visible.

The alternatives in Table 2.2 remain credible, for the reasons Section 1.5 set out. This workflow earns its place by making the support and approval boundaries explicit and open to inspection, not by being the best option for every reporting team (Gao et al., 2023c; Guo et al., 2024).

6.4 Public-web case study and future comparator

The public-web route extends the implementation from structured replay to bounded source discovery: web search proposes addresses, local controls decide whether a page may be collected, and a claim still needs a saved source, an exact supporting passage and approval (Gao et al., 2023a; OpenAI, n.d.-b). Controlled tests covered reviewed identity, network checks, request limits, write-once captures, redaction, exact passages, contradictions and approval using fake providers and fixtures, which establishes behaviour for those cases and nothing about the open web (Greshake et al., 2023; National Institute of Standards and Technology, 2023).

The most honest challenge to the whole custom build is that a managed platform might do it for less work. Grok Bot’s documentation describes persistent named Bots with defined jobs, tools, hand-offs and approval boundaries, covering a good deal of what Chapter 4 implements by hand (SpaceXAI, 2026). Lower effort remains a hypothesis, and the decisive question for any such comparison is whether the platform can refuse a claim the evidence does not support, because a configurable Bot is not by itself a verification contract (Dibia et al., 2024; Peffers et al., 2007).

6.5 Relationship to literature and design trade-offs

The findings agree with literature that separates factual correctness, citation support and source attribution. Exact locators and saved copies make a decision inspectable, but they do not establish source truth or complete coverage (Gao et al., 2023a;b). The observed D0 change is consistent with the purpose of a verification gate because the fixture was built to test that gate. It is not an independent confirmation of a general theory.

Fixed missing-state, cut-off and version rules also support repeatability. Their weakness is that a wrong or outdated rule repeats consistently. The rules therefore need named ownership, change control and re-evaluation when sources, schemas or reporting needs change (Gebru et al., 2021; Nikiforova et al., 2020; Pineau et al., 2021).

Human approval provides accountability for release, not automatic correctness. Interfaces should show uncertainty and allow correction, while studies of reliance show that people may still follow incorrect advice (Amershi et al., 2019; Bućinca et al., 2021). The pilot must measure reviewer edits, decisions and time instead of treating an approval button as evidence of benefit. Table F.6 in Appendix F separates the two answered research questions from the unexecuted comparisons. Table H.5 in Appendix H shows where the results support, qualify or leave the literature claims untested.

6.6 Generalisability and practical interpretation

Everything above applies to one local prototype, fourteen designed cases and a set of controlled security tests. Early-stage private-market evidence is unusually incomplete and sensitive to how terms are defined, which is precisely why statistical generalisation would need an independently selected and labelled set rather than a larger version of D0 (British Business Bank, 2025; Kaplan & Lerner, 2016; Kapoor & Narayanan, 2023).

Moving the design to another setting is not a matter of copying code. It would need a new data dictionary, a legal identity source, source licences, missing-value rules, a reporting cut-off, an evidence contract, a security boundary and a named approval owner. Checksums and versions help explain why two runs differ; they do not hold behaviour steady when the sources or models underneath change (Gebru et al., 2021; National Institute of Standards and Technology, 2023). The transfer checklist is Table H.4 in Appendix H.

The prototype is best understood, then, as a reporting control system that can be tested. It ranks nothing, predicts nothing and advises nobody, and whether it is genuinely useful remains a question for the pilot (Financial Conduct Authority, 2024; Hevner et al., 2004).

6.7 Prospective business pilot

The proposed pilot is a time-limited evaluation of the existing portfolio XLSX import as the named input integration. The prototype's Next.js review interface is the downstream report interface, with versioned JSON and accessible HTML exported only after approval. The pilot owner approves scope and the go/no-go decision. A reporting analyst prepares the manual baseline and operates each reporting case. A technical operator manages the local service, versions and failures. A reviewer or approver checks the evidence and current report version. A data and security owner approves data classes, source licences, storage, retention and any external-model route (Amershi et al., 2019; National Institute of Standards and Technology, 2023).

Before data collection, the team freezes the pilot's size as well as its rules. Four figures fix the size: how many portfolio companies, how many reporting periods, what calendar window, and the smallest number of completed paired reports that would make the comparison worth reading. The pilot owner and the reviewer choose those four; this dissertation leaves them open, because a size picked without knowing the site's caseload would be invented rather than planned. The team also freezes the eligible sources, schema, reference decisions, task completion rule and acceptance thresholds. It records setup effort in staff-hours for data mapping, source admission, environment configuration, training and dry runs. For every report it records analyst active minutes, reviewer active minutes, elapsed time, review and rework time, retries, model or API tokens, provider charges, source-licensing charges, storage and security costs. Cost per report is the allocated setup cost plus staff time, API use, licences, storage and security, divided by completed reports. No saving is assumed before this matched baseline exists (Peffer et al., 2007; Pineau et al., 2021).

Quality measures are supported-claim precision, evidence coverage, false acceptance, false rejection, reviewer edits, completion rate and source failures. Operational measures are active time, elapsed time and cost per report. The manual baseline uses the same inputs, source window and completion rule. Results are paired by company-period, and failed or abandoned cases remain in the denominator where the measure requires them. Data-quality measures must remain tied to the intended reporting use (Nikiforova et al., 2020), and reviewer disagreement must not be treated as proof that either person is correct (Artstein & Poesio, 2008).

The pilot proceeds only if authority, source licences, reference labels, storage controls, trained owners and rollback are in place. Before seeing outcomes, the pilot owner and reviewer set the minimum acceptable quality and completion values and the maximum acceptable time, false-acceptance, false-rejection and cost values. Time is on that list because a saving nobody committed to in advance cannot later be claimed as one. The recommendation is go only when those frozen criteria are met, approval cannot be bypassed, every exported claim has the required evidence record, and no stop event occurs. It is no-go or redesign when a criterion fails. This avoids inventing numeric targets in the dissertation while leaving a checkable decision rule (National Institute of Standards and Technology, 2023; Peffer et al., 2007).

The run stops immediately for unclear authority or consent, unexpected restricted or personal data, credential exposure, a source-licence or robots breach, external transmission outside the approved route, a broken evaluation freeze, evidence-record loss, approval bypass, uncontrolled cost or a security incident. The event, affected cases and recovery decision are recorded. Expected time or cost reduction remains a hypothesis until the paired pilot demonstrates it (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Four kinds of testing remain outstanding, and they are gathered here so that a pilot owner can find

them in one place. An independent tester needs to attack the collection and extraction route with hostile inputs. An accessibility assessment needs assistive technology and an assessor from outside the project. A security assessment needs to cover remote and multi-user operation, neither of which this prototype supports. And a live-source study needs to run the approved official connectors against real registry and publisher data. Sections 7.5 and 8.3 refer to this list rather than repeating it (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Ethics, Governance and Limitations

7.1 Data classification, privacy and purpose limits

At intake, evidence is marked restricted, internal, public or synthetic. Restricted portfolio and internal operating data stay local and cannot reach an external model. External processing of public or synthetic material requires explicit selection and a stated case purpose. Holding less data lowers the exposure, but the marking depends on a person getting it right, and it settles neither the legal basis for this work nor the university's approval of it (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Local source copies and exports are checksum-linked, written once where required, kept out of Git and privately permissioned, and personal contact details are removed before an accepted public page is stored. Two boundaries matter here. A checksum shows that a file has changed; it neither anonymises nor encrypts the contents. And information being publicly available does not settle whether this project may use it, because privacy, licence and purpose are separate questions from access (Central Digital and Data Office, 2023; Gebru et al., 2021).

Work with real restricted data or with participants stays held until an approved data-management and ethics plan names who controls the data and who owns purpose, access, retention, deletion and incident response. No retention period has been invented in the meantime. What the prototype demonstrates is a local purpose limit, which is a design property rather than a data-protection judgement (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

7.2 External-model processing and retention

The optional external route sends only approved public or synthetic company-level material and a reference that does not expose evidence. It requires open collection/model gates, reviewed identity, named reviewer, approved route and a runtime key. Requests use `store=False`, strict schemas and fixed limits. Restricted, internal or personal-contact content is prohibited, and model output cannot approve or export a report (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Live external-model requests did occur during development, and this section records them. On 26 August 2026, between 20:03 and 20:17 UTC, five bounded smoke runs sent six requests to the OpenAI Responses structured-extraction provider. Each run passed one checksum-pinned synthetic evidence item from the local fixture set through the ordinary extraction stage. No restricted, internal or real-

company material was sent, and the gates above were in force. Three requests succeeded on model `gpt-5.4-mini-2026-03-17`, each recording 449 input tokens, 79 output tokens and between 1.4 and 1.9 seconds. Three failed: one was refused by the provider, and two produced output that the local schema check rejected, the second after escalating to `gpt-5.4-2026-03-05` (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

These runs are development evidence about whether the route works. They are not a result. They sit outside the dated evaluation snapshot in Chapter 5, they measure no accuracy, latency or cost, and six requests on one synthetic item cannot describe provider behaviour. What the project can observe also stops at its own boundary: the runs confirm what was sent and what came back, but nothing about what the provider then retained. OpenAI separates saved application state from abuse-monitoring retention, and `store=False` is not Zero Data Retention, so deletion remains unverified. Before any use with real company material, the account settings, eligibility, current terms and data location need checking, and the confirming evidence needs recording (Central Digital and Data Office, 2023; OpenAI, n.d.-a).

7.3 Security and failure threats

Public pages may contain hidden instructions, misleading identities or malformed content. The route treats downloaded text as data, limits model tools, rejects instruction-like passages and demands exact support before it accepts a claim. The tests cover chosen risks; the controls cannot spot every hostile form, nor promise that a plausible supported statement is harmless (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Collection checks the protocol, embedded credentials, ports, network addresses, redirects, robots rules, content type, elapsed time and decompressed size, and connects to a verified public address while still checking the certificate hostname. Controlled tests covering internal-service access, address switching, redirects, content type and size all passed. Each test confirms that one specific input was refused, which is a narrower thing than a security assessment (National Institute of Standards and Technology, 2023; 2024).

Budgets, bounded retries, cancellation, task fingerprints, checksums and staged export together limit runaway cost, stale completion, altered files and partial writes. A forced stop can still leave work that needs deliberate recovery, and the dependencies and sources underneath can change without notice (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

7.4 Investment and public-person boundaries

The report links company events to evidence; it does not advise buying or selling, rank companies, create speculative valuations or assess suitability. Funding, awards, contracts, filings and attention are contextual and do not themselves cause or measure quality or future performance. Because financial information can influence decisions, external use requires separate review of purpose, audience and presentation (Estrin et al., 2024; Financial Conduct Authority, 2024).

The research concerns companies, not individuals. Captured contact details are removed; director, employee, investor or journalist profiles, inferred sensitive traits and reputation scores are excluded. A named public statement may support a relevant company claim but cannot become a judgement about that person. Human approval cannot expand these limits (Central Digital and Data Office, 2023;

National Institute of Standards and Technology, 2023).

7.5 Research and operational limitations

This section is the single place where the study’s limits are stated in full. Earlier chapters therefore report what was built and observed without repeating these boundaries, and a reader who wants the honest edge of every claim in this dissertation should read this section rather than hunt for caveats attached to individual results. Stating validity limits in one place, rather than scattering them, follows both design-science reporting practice and risk-management guidance that ties a claim to its tested context (Hevner et al., 2004; National Institute of Standards and Technology, 2023).

The principal evidential limit is D0 itself. Its fourteen cases are fictional, and they were written by the same person who wrote the rules they test, so the verifier difference may partly reflect that shared design rather than an independent property of the system. Behavioural test suites are built to expose specific known failures; passing them shows that the targeted behaviour is present, not that the system is sound in ways the suite never probes (Ribeiro et al., 2020). Each error category in D0 also has only one case, so no rate, distribution or confidence interval can be derived. Nothing in this dissertation estimates accuracy on real companies, parity with manual work, reviewer benefit or population performance (Kapoor & Narayanan, 2023; Pineau et al., 2021).

Several planned comparisons produced no observations and are absent rather than negative. The manual baseline, the participant-review condition, the live public-web study and the managed-platform comparison all required authority, frozen references or recorded human effort that was not available. Engineering evidence is bounded in the same way: 286 passing tests and 85.58 per cent statement coverage describe exercised local paths using fictional data and fake providers. Unit and end-to-end tests remain valuable, but in a changing world they are not sufficient evidence that a system works as intended (Pineau et al., 2021; Sculley et al., 2015). Construction and passing tests are therefore treated throughout as artefact evidence, never as effectiveness (Hevner et al., 2004; Peffers et al., 2007).

Anything outside the repository can move. Sources, licences, data formats, publisher pages, search ordering, model behaviour and provider policy all change independently of this code, and a checksum detects that a saved file has altered without preserving access to the original or its meaning. Fixed rules repeat consistently, which is a strength until a rule is wrong: manually set thresholds and frozen mappings become invalid as the world they encode moves on, and each one carries an ongoing maintenance cost (Gebru et al., 2021; Sculley et al., 2015). The researcher designed the system, wrote the cases and performed the scoring and interpretation, which is a clear confirmation risk. Any final evaluation therefore needs independently constructed references and a freeze agreed before the data are seen (National Institute of Standards and Technology, 2023).

Operationally the prototype is local, single-user, serial and SQLite-based. It has no production authentication, tenant separation, egress isolation, central monitoring, backup regime, service target or tested disaster recovery, and its accessibility was checked visually rather than with assistive technology or an independent assessor. The dated snapshot also retains five formatting failures and 22 deprecation warnings, and changes made after that snapshot are unevaluated. None of this cancels the controlled D0 result, but together these limits rule out any claim about usability, resilience, maintainability or production readiness. The end of Section 6.7 lists the testing that would be needed to close these gaps, and Table H.6, in Appendix H, summarises where authority and residual risk still sit (Amershi et al., 2019; Hevner et al., 2004).

Conclusion and Future Work

8.1 Answers to the primary question, RQ1 and RQ2

The primary question is answered within the controlled scope. The implemented workflow can preserve identity, reporting-period, missing-state, provenance, verification and approval records from intake to report export. Engineering checks and repository artefacts establish that these boundaries exist in the evaluated local version. They do not establish operational usefulness or production readiness (Hevner et al., 2004; Peffers et al., 2007).

For RQ1, the strongest result is traceable execution. D0 decisions can be followed from a fictional candidate and its evidence to the verifier state and report consequence, and the complete outcome list was identical across three repeats. The observed-zero, contradiction, stale-period and untrusted examples show what the evidence rules mean in practice. This answers how the implemented controls keep a report’s meaning intact within D0 (Gao et al., 2023a;b; Pineau et al., 2021).

For RQ2, separate verification changed claim admission under equal inputs. C1 emitted six candidates that D0 labelled unsupported, while C2 held all six and retained the five supported claims. Precision changed from 0.455 to 1.000, the unsupported-claim rate from 0.545 to 0.000, and recall remained 1.000. This is an observed verification-gate effect on fourteen designed cases. It is not evidence that multiple agents are generally superior or that the same rates will hold for real companies (Guo et al., 2024; Kapoor & Narayanan, 2023).

The dissertation’s conclusion is therefore narrower and stronger than the original research contract. It demonstrates an evidence-first implementation and one controlled verification effect. Manual comparison, participant review, live source use and business benefit are outside the active questions because the required observations were unavailable. Their absence does not make the executed result incomplete; it sets the boundary for future evaluation (Hevner et al., 2004; Pineau et al., 2021).

8.2 Demonstrated contribution

The main contribution is a reporting workflow that makes source admission, claim support, conflicts, missing information and release authority explicit. Separating the roles gives production, verification and approval different permissions and saved hand-offs. The design is inspectable because the saved records link each decision to a source, rule, version and output (Buneman et al., 2001; Pineau et al., 2021).

The empirical contribution is the paired D0 comparison. It reports aggregate metrics and exact case

examples, including a correct accepted zero and three different reasons for holding a claim. The result supports the verification contract on the fixture rather than a new algorithm, a persona effect or a claim of production performance (Gao et al., 2023a;b).

The practical contribution is an executable pilot design. It names the existing XLSX input and Next.js review interface, assigns staff responsibilities, defines cost and time records, specifies quality and operational measures, and sets go/no-go and stop rules. Time saving and cost reduction remain hypotheses until that pilot is authorised and run (National Institute of Standards and Technology, 2023; Peffers et al., 2007).

8.3 Sequenced future evidence

The first next step is the prospective business pilot in Section 6.7. It should freeze the data, source window, reference decisions, completion rule and acceptance criteria before collecting paired manual and workflow observations. The pilot must record active and elapsed time, reviewer work, supported-claim precision, evidence coverage, false acceptance, false rejection, source failures, completion and cost per report. A pilot result should not be used to tune a sealed final evaluation set (Kapoor & Narayanan, 2023; Pineau et al., 2021).

A separate live-source study can then compare approved official connectors with bounded assisted discovery using identical companies, dates, source limits and output requirements. Any external model route requires current provider controls, approved data classes and recorded usage. The outstanding testing listed at the end of Section 6.7 must be finished before remote or multi-user operation is considered (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

A managed platform such as Grok Bot should then be tested against the custom orchestration layer on the terms set out in Section 6.4. The question is whether a bought platform can reproduce the same evidence-admission, verification and approval contract for less setup and maintenance effort, measured under equivalent inputs rather than assumed from its feature list (Guo et al., 2024; Pineau et al., 2021; SpaceXAI, 2026).

The full evidence roadmap is Table H.7 in Appendix H. Each future stage requires an explicit authority and stop decision. None is presented as a result of this dissertation (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

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Evidence of Required Ethics Training

Candidate-supplied evidence is pending and must be inserted before submission.

Ethical Approval or Waiver Confirmation

Candidate-supplied approval or waiver evidence is pending. The corresponding pro-forma box and ethics reference remain deliberately blank until this evidence is inserted.

Requirements and Implementation Traceability

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Local runtime and accountable actor NFR-SEC-001; FR-UI-003-004	One configured local operator uses the loopback Next.js dashboard, private FastAPI service, SQLite and ignored local files.	Next.js is the only published browser surface; its same-origin server-side proxy reaches FastAPI privately. Host/client checks, server-held CSRF credentials and configured reviewer identity protect changes.	Formal accessibility/security review and any wider deployment design require separate evidence and authority.	No anonymous, remote or multi-user access. Reject non-loopback clients, unexpected hosts, missing CSRF or missing configured reviewer.	<i>web.py</i> ; dashboard proxy; Compose and route tests. IN PLACE Local only; not production sign-in.
Identity-first intake FR-ID-001-004; FR-INT-001, 006	A submitted Companies House number is an identity claim; a named reviewer may accept or reject it with rationale.	A number with a valid format creates an unresolved company and unreviewed identifier. Names and domains do not merge records automatically. Identity conflicts and data-class mismatches stop the case.	Before direct live registry use: close G2 with current interface/terms, credential, identifier, storage, cutoff and attribution review.	Syntax is not legal-identity authority. Do not collect or link facts while identity is unresolved, conflicting, rejected or from another classification boundary.	Intake service; ADR-0007; tests. IN PLACE Offline; live registry held.
Data class and authority NFR-SEC-002-005; NFR-RES-001	Restricted and internal material stays local. Accepted public evidence and fictional cases follow separate routes.	Fixed data classes; purpose and person records; write-once files with private “0600” permissions; records without content; provider check.	Verify lawful authority, controller, access, retention and deletion, and any ethics/consent terms before restricted-data or participant work.	Never place restricted or internal content in Git, fixtures, raw logs or an external model. Unclear authority or unexpected personal data stops processing.	<i>enums.py</i> ; <i>security.py</i> ; boundary tests. CONTROL IMPLEMENTED ; authority not verified.
Public and model boundary FR-EXT-004-008; FR-RES-001-004	Only a public case with a reviewed company number and cut-off may enter limited web search, collection and optional extraction. Fictional cases test the route offline.	External and collection gates start closed. Snippets and model text are suggestions, not evidence. Protected HTTPS collection, rejection of hidden instructions, “store=False”, strict passage and schema checks, and separate verification are required.	Separately authorise and evaluate the live public company path; finish source-specific G2 review and verify provider controls at execution. Direct registry execution remains P1.	No restricted or internal model input, person profiling, arbitrary browsing, snippet support, autonomous recommendation or model approval/export. Missing, held or conflicting evidence cannot become support.	<i>bootstrap.py</i> ; ADR-0008; controlled boundary tests. CODE AND LOCAL TEST EVIDENCE ; live company route not run.

Continued on next page

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Review and export FR-VER-002; FR-REP-004-009	The configured reviewer assesses a versioned report or profile with verification records, then approves, rejects or returns it.	Person, reason and version record; version lock; edits remove approval; no export before approval; files checked against a list and hashes before reuse.	Authorised C0/C3 observation and reviewer utility remain pending; production roles or signed approval need a separately scoped design.	No model or workflow stage can approve or publish. Unverified claims, stale versions, rejected reports, missing approval, tampering or incomplete writes stop export.	<i>reporting.py</i> ; approval and export tests. IN PLACE Offline; human result unavailable.
Prototype exclusions Charter P0/P1/P2	Engineering evaluation of controlled intake, evidence, verification and local review for RQ1 and RQ2.	Fixed rules for each stage, clear failures, source records and local audit make unsupported actions visible instead of quietly expanding scope.	Live connectors, participants, restricted-data study or deployment may proceed only through named future gates; they are not completion criteria for this slice.	No supplied credential use, production tenancy, social scraping, unrestricted crawling, autonomous publication, investment decisions, fine-tuning, cloud deployment or native PPTX/PDF generation.	Project Charter “Explicit non-goals”; requirements P1/P2; architecture gaps. CONTRACT BOUNDARY ; not an empirical result.

Source note: Developed by Faye Niu from versioned repository requirements, accepted design decisions and local checks. NIST (2023, PDF pp. 32–35) supports documented scope, oversight and independent assessment; Greshake et al. (2023, PDF pp. 1, 3–5) supports treating retrieved content as an untrusted instruction boundary; and Amershi et al. (2019, PDF p. 3) supports clear capability limits and efficient human correction. These sources do not certify this system. The matrix is a design summary, not a study result. Built means named code plus local checks, not live-source completion, ethics approval, restricted-data authority, security effectiveness or production readiness.

Table C.1. Prototype requirements, controls and limits

Connector / source	Approved function / version	Identity, cut-off and period rule	Source location and saved copy	Outcome states	Current status / limit
Connector ability check	Source list plus policy version; exact rules for identifiers, facts, events, file types and collection modes. Derived-data rule <i>v2</i> .	Reviewed source-scoped identifier must belong to the company and be valid at cutoff. Cumulative facts require programme start through cutoff.	A fingerprint links the source and version, identifier, cut-off, programme window, requested facts and mode. Source-copy and derived-data SHA-256 values are checked on replay.	Before a call, blocked or invalid is different from a saved record , no-record , unavailable or failure . A rule-version mismatch stops the call.	BUILT TESTED OFFLINE. A traceable and consistent run does not prove the source is true.
Companies House replay	Companies House source; source list <i>1.4.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact reviewed Companies House number. Filing, charge, incorporation and dissolution events must occur by the cut-off. Account states remain separate.	Write-once test bytes, checksum and source version; exact JSON locations and stable filing and charge identifiers; versioned derived-data hash.	Record; exact-ID no-record; unavailable source copy; invalid ID, duplicate or rule set; final failure kept separately. A blank is never changed to zero.	BUILT FICTIONAL REPLAY. No live registry result, current legal truth, complete coverage, valuation or production claim.
UKRI replay	UKRI source; source list <i>1.3.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact UKRI organisation ID. Latest version available by cutoff is used. Cumulative totals require the exact programme-start–cutoff window.	Write-once test checksum; stable project and outcome-ID locations; the derived-fact location records the period, included projects, method and schema version.	Record or exact-ID no-record; invalid scheme, period or rules; failure remains separate. Missing or non-GBP amounts create clear incomplete states, never a zero or complete total.	BUILT FICTIONAL REPLAY. No live UKRI access, complete coverage, cause-and-effect or source-truth claim.
Companies House API mode	Read-only API mode appears in source list <i>1.4.0</i> , but its live-use flag is false. Licence and terms evidence is on hold.	An exact reviewed number, company link and cut-off would still be required. An API key or valid number format would not give permission.	No approved live copy exists. A future run would need write-once response bytes, source details, exact locations, and reviewed evidence for the source and version.	Blocked before connector invocation. Invalid configuration or identity remains invalid; no live record, no-record, unavailable or failure evidence exists.	HELD LIVE-UNRUN. Code presence is not permission, official retrieval evidence, availability or production readiness.
Public-web discovery / capture	Separate public-web research <i>v1</i> : find possible web addresses and collect pages safely. It is not a registry connector or a replacement for official source approval.	Public case, reviewed Companies House number, fixed cut-off and agreed claim or measure scope. Downloaded text stays outside the instruction boundary.	Search output is only a list of possible sources. Evidence needs accessible captured bytes, checksum, publisher and web address, exact passage and a source-specific decision.	Possible, collected, blocked, unsupported or failed gaps remain clear. Missing, held or conflicting evidence cannot become support. No search result is not an official no-record finding.	CONTROLLED ROUTE NOT USED IN A LIVE STUDY. Stop controls and local checks do not prove coverage, source truth, quality or live availability.

Table C.2. Fixed source connectors and their current states

State rule: “record” means a successful collection containing an accepted record; it is not a truth label. No-record, unavailable, blocked, invalid and failure retain different meanings and reporting consequences. *Source note:* Developed by Faye Niu from versioned repository rules and local fictional tests. Nikiforova et al. (2020, PDF pp. 1–3, 15–17) supports use-case-specific quality checks; Gao et al. (2023a, ALCE, PDF pp. 3–4, 10) separates traceability/citation support from answer correctness; NIST (2023, PDF pp. 33–35) supports documented test conditions, uncertainty and limits. These sources do not certify a connector, publisher record or live-source result. The matrix is a design summary, not a coverage, quality, availability or production result.

Area	Failure kept visible	Recovery or control	Evidence available
Identity and intake	Invalid, ambiguous or conflicting identifier; duplicate contract disagreement.	Named resolution; exact compatible repeats reuse the existing case.	Implemented; controlled tests. No live registry truth.
Fixed sources	Policy block, no record, unavailable source, invalid rules and final failure remain separate.	Offline checksum replay; stop on an unsupported function, identity, cut-off or version change.	Built; fictional replay. Direct live sources held.
Public-web research	Search, collection, extraction or report writing may be blocked, cancelled, reach its limit or fail.	Limited retries and use, stop controls, fixed task fingerprint and recovery of interrupted work.	Built; controlled tests. Live check and fixed comparison not run.
Claims and conflicts	Unsupported, old, untrusted, insufficient or conflicting evidence is not accepted.	Exact passage and source check; separate status; named decision on conflicts.	Built; fictional and controlled tests. Real-company effectiveness not measured.
Review and export	Old version or hash, incomplete verification, changed content or partly written output blocks release.	Named decision, removal of approval after an edit, version lock, file-list and hash checks.	Built; controlled tests. No automatic or remote publication.
Evaluation and operation	C0 and C3 observations, live company results, production availability and disaster recovery are absent.	Keep blank and held states. Require separate permission and a fixed study plan before running them.	Planned or outside scope. No outcome claim.

Table C.3. Failures, recovery controls and current status

Source note: Developed by Faye Niu from the repository requirements, design, evaluation plan, source register and controlled test evidence at the Section 5 snapshot. Status describes executed engineering evidence only; it is not a production, availability, source-truth, user-benefit or investment-performance result.

Dataset, Source, and Freeze Manifests

Condition	Workflow	Independent verifier	Human involvement	Scoring point	Fair-comparison controls	Current evidence state
C0 Manual	An authorised practitioner completes the current workflow in the fixed information window. Every tool or model used is logged.	No condition-level verifier.	Practitioner produces the report; there is no later C3 review stage.	Completed manual report; timing and corrections only if authorised.	Same measure definitions, period, eligible input, source-access window and agreed reference answers.	PLANNED / ON HOLD Waiting for permission, reference answers and a fixed study plan.
C1 Fixed single-agent	One extraction and report-writing route uses the shared catalogue and cleaning rules. Source records are kept.	No.	No human edits before scoring.	Automated C1 report before any review.	Shared controls at left, plus fixed schemas, prompts, code, environment and stop rules.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C2 Controlled multi-agent	Separate roles collect, extract and clean data. The verifier checks support before fixed report writing.	Yes: separate from extraction and composition.	No human edits before scoring.	Verifier-gated C2 report before human C1 review.	Same shared and fixed technical controls as C1.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C3 C2 + HITL	C2 output enters the versioned named-review workflow.	Yes: inherited from C2.	Named reviewer may edit, approve or reject under the authorised protocol.	Approved post-review version, scored separately from pre-review C2.	Same shared controls and C2 set-up, plus the fixed review plan.	WORKFLOW BUILT Human results are planned and on hold.

Table D.1. C0–C3 evaluation-condition matrix

Fair-comparison rule: conditions use the same measure definitions, reporting period, eligible input, source-access window and agreed reference answers. Any tool or model used in C0 must be recorded. *Exclusion:* the public-web company-research route has not run live. It is not a C0-C3 condition and provides engineering evidence only. *Status:* developed by the author from the fixed project rules, with no study results or pass thresholds.

Stage	Required action	Required record or output	Reason to hold the study
1. Define Codebook + source policy	Create a versioned labelling guide from the data dictionary, with examples for measures, claims, sources and periods. Define which sources may be used and the cut-off date.	Labelling-guide version and hash; label schema; source rules and source-list template.	Hold labelling and scoring until the scope, examples, source rules and cut-off are clear.
2. Label Work separately	Two authorised reviewers with relevant subject knowledge label the same fixed set of sources independently. Keep their records separate until agreement is measured.	Reviewer IDs; source-set version; separate structured labels and times.	Unauthorised reviewer, different source set or discussion before the separate labels are complete: HOLD .
3. Record Complete label	Record the agreed value or reason it is missing, whether evidence is eligible, whether a source can be found, the period, claim support and required report section.	Structured label with source and location, plus a clear missing, unclear or conflict state.	Good writing or a citation does not prove support. Unsupported or unclear items remain held.
4. Measure Before discussion	Report raw agreement. Add a statistic corrected for chance only where the sample, label scale and task design support its use.	Total number of eligible labels; direct agreement; suitable adjusted measure, assumptions and uncertainty; otherwise NULL .	Do not import a pass threshold. High direct agreement can hide an uneven label balance, bias or differences in how text was divided.
5. Agree final labels Give reasons	Compare each disagreement with the fixed passage. Record the reason for the final choice and any change to the labelling guide.	Final label; reason; type of disagreement; revised guide where needed.	No silent change, no assumption that a majority is true, and no change to a definition without a new version.
6. Fix Reference + source list	Fix the final reference answers, labelling guide, source set and source-list hash before scoring any condition.	Write-once reference ID, version and hash, plus the scorer's input list.	Manual or system output is never assumed to be true. A later change needs a new version and a record of the change.
7. Score Blind where practicable	Score conditions against the same fixed reference without showing the scorer the condition where practical. Record any case where the condition becomes known.	Scores for each condition linked to the reference; record of blind scoring and any departure from it.	C0 manual and C1-C3 system outputs are conditions being tested, not the reference answer.
8. Keep equal Sources + time	Give conditions the same definitions, period, eligible source window and reference answers. Record inaccessible, deleted, paywalled, sign-in-only and later sources. Keep publication and availability separate from collection time.	Company-period source list; source state; publication, availability, collection and cut-off times; any difference in access.	Unequal access or support after the cut-off: hold the claim and report the difference. Never change missing evidence to zero.

Table D.2. Agreed reference answers and equal source access

CURRENT STATUS: REFERENCE ANSWERS HAVE NOT BEEN CREATED OR FIXED; THERE IS NO STUDY SCORE. *Source note:* Developed by the author from the current evaluation plan and data dictionary, with locally stored Artstein & Poesio (2008, PDF pp. 2–8, 16–23, 36–37), Gao et al. (2023a, ALCE, pp. 3–5, 8–10), Pineau et al. (2021, pp. 1–5, 9–13), Gebru et al. (2021, pp. 2, 4–10) and NIST (2023, pp. 32–35). This is a plan, not a result. It does not supply a result value, sample size or pass threshold.

Source Admission and Provider Data Controls

Risk	Control	Record kept	When to stop or ask for help	Current status
Restricted, personal or secret data is sent without permission	Classify data at intake and again before an external provider. Keep restricted and internal material local and out of external processing. Use private file permissions and record IDs, counts and hashes rather than raw values.	Data class, purpose and person responsible; source-copy and export checksum and file list; review decision; record without content; security check.	Unclear authority, unexpected personal data, an exposed credential or an unapproved processor: stop collection, isolate and record the event, then seek supervisor, ethics or security direction.	CONTROL BUILT / TESTED. Permission for real data, retention and deletion times, and confirmation that exposed keys were changed are NOT VERIFIED . No restricted external run is claimed.
Unauthorised C0 or C3 observation or reviewer feedback	Require recorded ethics and data permission, consent and withdrawal rules, approved participant information and questionnaire, coded IDs, minimum data and a quotation plan set in advance.	Approval or waiver; dated information and consent versions; separate ID link; task, review and withdrawal records; hashes before and after review.	If approval, consent or withdrawal handling is missing or unclear, do not recruit, observe or collect. Keep the result blank and ask for guidance.	APPROVAL NOT VERIFIED / ON HOLD. Human C0 and C3 work and feedback are unavailable. No participant result is claimed.
D0, D1 or D2 information leaks across study stages	Accept D0 by hash and test name. Use D1 only after permission. Keep D2 sealed with no visible path until the evaluation choices are fixed. Keep company and period groups separate. A change based on D2 makes the run exploratory.	Dataset list, hash and source version; group and period split; code, rule, prompt, model and analysis versions; loader result; access log and changes from the plan.	If the seal, checksum or group split fails, or D2 is viewed or used for tuning, stop the final study. Record the change and ask the supervisor to review it. Any later run is exploratory.	D0 CONTROL BUILT / TESTED. D1 has not run. D2 is sealed and blocked by the loader. No D1 or D2 result is claimed.
Public source is used unlawfully, is unavailable or is too late	Use versioned source rules, reviewed identity and cut-off, GET-only access with clear limits, robots and access checks. Do not bypass forms, sign-in, paywalls or CAPTCHAs. Review licence and sharing rules for each source.	Source ability and terms with review date; request fingerprint; availability, collection and cut-off times; copy hash and exact location; inaccessible or held state.	Unknown or changed terms, forbidden access, after-cutoff support or redistribution uncertainty: hold the source or claim; do not fetch or export; require named review.	LIMITED CONTROLS BUILT. Local public-web use is conditional. Direct live registry evaluation and sharing remain on hold or unavailable.

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Risk	Control	Record kept	When to stop or ask for help	Current status
Prompt injection or unauthorised model use	Treat downloaded text as data and reject content that looks like instructions. Allow only chosen public or synthetic input, set “store=False”, require a strict schema and exact passage, limit models and attempts, and never give the model approval or export rights.	Data class; provider, model and schema versions; attempt, token and error logs; exact passage and verification; reason for rejection or no claim; file list without content.	Injection, personal or restricted content, arbitrary routing, unclear processor authority or failed span validation: reject or abstain before claim support; stop model use and export.	CONTROL BUILT / TESTED. D1, D2 and participant processing have not run. There is no result for model quality, provider retention or better performance.
Version or setting changes; incomplete record	Fix the code revision and changes, dependencies, database version, source and dataset lists, cut-off, catalogue, labelling guide, prompts, schemas, connectors and models. Remove secrets. Keep run and agent IDs and hashes for analysis and exhibits.	Repeatability record; write-once input and result hashes; environment and database version; attempts, failures and human decisions; changes from the plan; hashes for generated files.	A missing or mismatched hash or version, secret in the set, unexplained change or failed repeat stops the comparison. Keep the failure and resolve it before making a claim.	MECHANISMS BUILT / TESTED. D0 repeat evidence exists. There is no final study record because the held conditions have not run.
Reference bias; builder and evaluator bias; selective reporting	Plan a versioned labelling guide, two authorised people working independently and a recorded final decision. Keep scoring blind where practical. Set outcomes and calculation totals in advance. Keep exclusions, failures, blank and negative results, and limit conclusions.	Separate labels; agreement, final decisions and labelling-guide versions; reference and scorer hash; case flow; analysis, changes from the plan and limits.	No authorised fixed reference, a rule changed after seeing outcomes, an unavailable calculation total or an unresolved disagreement means no final score. Mark the run exploratory or held and seek supervisor and statistical review.	DOCUMENTED PLAN. Reference answers and the final study plan are not fixed. C0, C3, D1 and D2 have no study results.

Status key: **built/tested** means repository code plus selected local checks run the stated mechanism; **documented plan** does not mean it was run; **approval not verified** means no claim of approval or refusal is made; **unavailable/sealed** cannot supply a result. *Source note:* Developed by Faye Niu from the current draft evaluation plan, governance rules and repository contracts, using local NIST (2023, PDF pp. 32–35), Gebru et al. (2021, pp. 2, 4–10), Greshake et al. (2023, pp. 1, 3–5), Kapoor & Narayanan (2023, pp. 1–7, 17–21) and Pineau et al. (2021, pp. 1–5, 9–13). This is a plan, not a result. The controls reduce risk but do not prove ethics compliance, lawful source use, security, repeatability, correctness or effectiveness.

Table E.1. Ethics, leakage and repeatability controls

Supplementary Evaluation Results

Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Claim precision	Scored emitted claim	Supported emitted claims.	All emitted claims eligible for scoring.	No eligible emitted claims: NULL / N/A with reason, never zero.	Treat claims from the same company or report as related; interpret with recall.
Claim recall	Required reference claim	Required reference claims produced correctly.	All reference claims required by the fixed scope.	No required labelled reference claims: NULL / N/A , never zero.	Group by company and report; missing claims remain visible.
F1	Condition + fixed claim set	$2PR/(P + R)$, using the precision and recall above.	Same fixed scoring set; no separate total.	NULL unless P and R are defined and $P + R > 0$.	Report P and R separately; F1 is not an acceptance threshold.
Unsupported-claim rate	Emitted claim	Produced claims without reference or evidence support.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Fix the support rules before scoring. Good writing or a citation does not prove support.
Claim-support rate	Emitted claim	Produced claims that meet the fixed support rule.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Keep separate from answer correctness and complete source records.
Contradiction recall	Reference conflict	Reference conflicts flagged correctly.	All reference conflicts.	No labelled reference conflicts: NULL / N/A , never zero.	Group linked conflicts; report false conflict flags separately.
Identity, acquisition, temporal + source quality	Company–source–period layer	Correct identity, acquisition state, temporal eligibility and source-quality outcomes, each separate.	TO DECIDE BEFORE STUDY: units with labels for each layer.	A layer with no labels is NULL ; event outcomes are NULL in current D0.	No single quality score; group by company, report and whether a source can be found.
Schema + extraction	Extraction record / field	Output that follows the schema; exact or field-level match to the fixed label.	TO DECIDE BEFORE STUDY: attempted and eligible records, and record-level or field-level scope.	No labelled eligible records: NULL , never zero.	Report by format, measure type and whether a source can be found. Fields in one report are related.
Normalisation + missing state	Company–metric–period field	Correct type, value, unit or currency and correct missing-state label; retain blank–zero matrix.	TO DECIDE BEFORE STUDY: labelled fields in scope by task and data type.	No labels: NULL . Zero is a valid observation, not missing; zero accuracy requires a positive denominator.	Report by data type/missing class; do not pool repeated company fields as independent.
Complete source records	Eligible emitted claim	Claims with source ID, locator, applicable publisher, retrieval time, checksum and period.	All eligible emitted claims.	No eligible emitted claims: NULL / N/A , never zero.	Completeness makes a claim traceable. It does not prove that the source or claim is true.
Source-location validity	Claim and location in a fixed source copy	Exact location opens the accepted fixed source copy.	TO DECIDE BEFORE STUDY: eligible locations and claims, and how several locations are handled.	No eligible inspected locators: NULL ; a broken inspected locator is a failure, not missing data.	Keep resolution separate from passage support and source truth.

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Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Report-section completeness	Report + required section	Required section status against the fixed reference template.	TO DECIDE BEFORE STUDY: required sections and how held or empty sections are handled.	No fixed template or scored report: NULL .	Aggregate at report/company level; presence does not establish factual correctness.
Factual errors + severity	Scored claim / report	Error count using categories set in advance; seriousness label for each error.	Count per report; the rule for combining seriousness is TO DECIDE BEFORE STUDY .	No scored report: NULL . Zero only after an eligible report is checked and no error is found.	Fix seriousness definitions before the scorer sees the condition or outcomes.
Transparency rate	Missing, conflicted or stale item	Eligible items surfaced with their state and reason.	TO DECIDE BEFORE STUDY: exact eligible items and how report-level results are combined.	No eligible labelled items: NULL / N/A , never zero.	Report missing, conflict and old-information groups separately. Do not use one score without context.
Human time	Authorised human + report	Active human minutes and elapsed wall-clock minutes, recorded separately with common start/stop rules.	Match the same company, period and report under one “review-ready” definition. How results are combined is TO DECIDE BEFORE STUDY .	Current D0 and unauthorised C0/C3: NULL ; never reconstruct missing time as zero.	Group repeated tasks by reviewer and report. Record tools, interruptions and waiting.
Machine stages + retries	Stage, attempt and run	Stage duration, attempt/retry and failure/re-run events.	Executed stage or run; summary method and a comparison that includes failures are TO DECIDE BEFORE STUDY .	Unexecuted stage: N/A with reason, not zero.	Pair same company-period; retain failures and stage boundaries.
Tokens + monetary cost	Model, attempt and run	Input/output tokens by model/run; cost from official execution-date price, recorded formula and currency.	Same executed workflow unit and model/version; do not estimate an absent price.	Missing logs or price: NULL . Zero only when logs prove there was no billable use.	Report tokens separately from cost; version and date bind the comparison.
Repeat reliability	Fixed configuration + repeated unit	Complete equality for fixed outputs; field or claim agreement for settings that can vary; completion, schema validity and failure categories.	TO DECIDE BEFORE STUDY: repeat set, output scope, matched total and agreement measure for settings that can vary.	No qualifying repeats or denominator zero: NULL , never zero.	Group by company and report. Show failures and configuration hash. There is no universal pass level.
C3 usability + edit burden	Authorised reviewer + task/report	Ethics-approved task success, time, changed decisions, whether confidence matches correctness, or a licensed scale; edit events and optional text or meaning-based distance.	TO DECIDE BEFORE STUDY in the authorised questionnaire and task plan. Keep C2 before review and C3 after review.	Current D0 with no participant observation: NULL . Do not invent reviewer value.	Group by reviewer and report. Record order and experience. Do not choose a scale after seeing outcomes.
CURRENT STATUS: PLANNED. THERE IS NO STUDY RESULT OR PASS THRESHOLD.					
Universal rule: no eligible labelled cases or a zero denominator yields NULL / N/A with a reason , never a measured zero. <i>Source note:</i> Developed by the author from the current evaluation plan and data dictionary, using local Gao et al. (2023a, ALCE, PDF pp. 3–4, 10), Nikiforova et al. (2020, pp. 1–3, 15–17), Pineau et al. (2021, pp. 1–5, 9–13), Artstein & Poesio (2008, pp. 2–8, 16–23, 36–37), NIST (2023, pp. 32–35) and Amershi et al. (2019, pp. 3–5, 16–19). These sources motivate separate and documented measures that include human work. They do not validate this system or supply its thresholds.					

Table F.1. Measures, calculation totals and interpretation rules

Case	Category	Candidate	Evidence state	Expected decision	C2 consequence
complete_internal_support	Complete	Employees = 12	Matching submission	Supported	Emit
sparse_not_reported	Sparse	Not reported	No evidence required for a claim	No claim	Hold
zero_is_observed	Blank / zero	Jobs = 0	Matching submission zero	Supported	Emit
blank_is_missing	Blank / zero	Empty value	No evidence	No claim	Hold
mixed_type_integer	Mixed types	Employees = "17"	Matching integer 17	Supported	Emit
conflicting_public_value	Conflict	Grant = GBP 100	Same-period evidence = GBP 200	Contradicted	Hold conflict
stale_public_value	Stale	Awards = 1	Matching value in prior period	Stale	Hold
unsupported_no_evidence	Unsupported	News items = 4	No evidence	Insufficient evidence	Hold
ambiguous_company	Identity	Employees = 8	Matching submission, identity unresolved	Identity hold	Hold
duplicate_same_value	Duplicate	Grant = GBP 300	Two matching sources	Supported	Emit
inaccessible_source	Inaccessible	Grant = GBP 700	Source unavailable	Insufficient evidence	Hold
narrative_structured_fact	Narrative	Products = 2	Matching narrative evidence	Supported	Emit
mixed_currency_conflict	Mixed currency	Investment = GBP 400	Evidence = USD 400	Contradicted	Hold conflict
prompt_injection_evidence	Prompt injection	Partnerships = 5	Matching value marked untrusted	Rejected untrusted	Hold

Table F.2. Complete D0 fictional-case decision ledger

Source note: Exact case identifiers, categories, expected states and values from `fixtures/evaluation_cases.json`. The ledger is synthetic and does not describe any company.

Area checked	Cases exercised	Result	Limit
Full research route	Reviewed identity, limited discovery, capture, exact passage, v3 report, named review and download	PASS	Fake or local evidence; no result for live accuracy, coverage or usefulness.
Task and budget rules	Fixed fingerprints, call budget including retries, time limit, zero redirects, cancellation and recovery after interruption	PASS	No live result for delay, tokens, provider limits or cost.
Network protection	HTTPS, credential and port checks; DNS changes; redirects; robots rules; file type and size limits; connection pinning and TLS	PASS	Selected threats only; not a complete server-side request forgery or production security test.
Evidence checks	URL matches the source list, run ownership, duplicate removal, exact passage and supported-value checks	PASS	A supporting passage does not prove publisher truth or complete discovery.
Source mix and conflict	Sources rotated by publisher, balanced batches, and preserved contradictions and gaps	PASS	No estimate for real publisher coverage or recall.
Review and cleanup	Old or changed decisions rejected, source ownership checked, and one run cannot clear another	PASS	No result for reviewer benefit or several users working at once.
Focused suite total	37 tests; one dependency deprecation warning	37 PASS	Controlled local execution only; all external calls absent.

Table F.3. Controlled company-research and hostile-input results

Source note: Fresh local execution of company-research integration, database update and fictional-case tests. PASS means that the named check passed. It does not approve a source, model, company profile, human process, network boundary or production deployment.

Comparison element	Required fixed evidence	Current state	What can be reported
Units and identity	Authorised public-company list; reviewed Companies House number; company cut-off date	HELD	No sample size, company result or wider claim.
Equal source access	Eligible-source list, availability states and the same cut-off window	HELD	No fair comparison between official-only and assisted routes.
Official-only condition	Approved live official connectors and write-once captures	HELD	Fictional replay cannot replace live official evidence.
Assisted condition	Authorised count, checksum and locator check; fixed query, model, prompt and budget; protected capture	BUILT; NOT RUN LIVE	No live discovery, delay, token or cost value.
Independent reference	Relevant claims, exact passages, conflicts, gaps and unavailable sources, fixed before scoring	NOT CREATED	Precision, recall and category coverage remain unavailable.
Planned measures	Support rates, source records, conflicts, source and category coverage, gaps, time, retries, tokens and cost	PLANNED	No value is reported because no suitable total exists.
Interpretation	Paired comparison for the same company and cut-off, with stated budgets, missing data and failures	FUTURE	It could show limited added discovery, but not a complete web search or investment quality.

Table F.4. Planned public-company source-discovery comparison

Source note: Author summary of the fixed study plan, approved-source list and current implementation. This table describes future work. It contains no result for a company, model, source search, cost, time or usefulness.

Condition	Built or planned	Missing permission or evidence	Current result
C0 manual	Planned same-source period, clear stopping point, active and total timing, and a plan to record tools and corrections	Authorised practitioners, ethics and data permission, fixed tasks, independent reference answers and sample basis	0 cases; all quality, time, cost and usefulness measures NULL .
C2 before review	Fixed verified output can be saved and scored before a person sees it	No authorised real-company or reference dataset	D0 mechanism results only; no human comparison result.
C3 HITL	Versioned interface, edit history, approval, rejection and reasons	Authorised reviewers, consent and study form, tasks hidden from the reference answers, order plan and observed events	0 cases; edits, reversals, timing, quality and reviewer usefulness NULL .
Reference process	Independent labels from two reviewers, agreement, a way to settle differences and a fixed version are planned	No completed labels, observed disagreements or final decisions	Agreement and reference-quality measures NULL .
Future-pilot consequence	The review design and planned paired measures can be described	No human observations	No observed claim about time saving, error correction, usefulness or decision quality.

Table F.5. Manual and human-in-the-loop comparison status

Source note: Author summary of the fixed study plan and current evaluator results. The table reports that the human study was not run. It does not show a negative effect, and no participant data were collected or inferred.

Question	Available evidence	Answer supported by the evidence	Missing evidence or limit
Primary question	Implemented local workflow; engineering tests; D0 C1/C2 comparison	ANSWERED WITHIN CONTROLLED SCOPE: evidence records and the verifier effect are demonstrated	No inference to real-company accuracy, business benefit or production use.
RQ1	Identity, time, source, verification and export controls; fourteen designed D0 cases; three repeats	IMPLEMENTATION AND CONTROLLED BEHAVIOUR: the required states and links are retained	No estimate for real-world accuracy or source coverage.
RQ2	Paired C1/C2 D0 output; five supported and six unsupported emitted candidates; zero local LLM cost	OBSERVED D0 EFFECT: the verifier held the six designed unsupported candidates without reducing D0 recall	Visible labels helped shape the test, false rejection outside D0 is unknown and timing is not useful.
Future pilot	Manual baseline and versioned named-review controls are defined	NOT A CURRENT RESULT: the design is prospective	No participant observations, active time, cost, editing or usefulness evidence.

Table F.6. Research-question evidence and answer status

Source note: Author summary of Chapters 1, 3 and 5. The status labels separate what was built, what was measured with fictional cases and what is proposed for the pilot.

Reproducibility and Build Manifest

Record item	Recorded value	Meaning
Host/runtime	Darwin 25.6.0 arm64; Python 3.12.13	Local execution environment; not a portability result.
Git state	HEAD 6356aa49cb3b; dirty working tree recorded	The commit is a base reference, not a complete content seal.
Python dependencies	pyproject.toml SHA-256 d8ed0555...; runtime lock 5cce8f8e...; development lock 5e2469bc...	The declared and installed dependencies are recorded with checksums.
Database	Alembic revisions 0001–0009; current head 0009	Fresh SQLite upgrade and schema-drift check executed.
Evaluation list	SHA-256 1c16234a...; schema evaluation-manifest-v2	D0 was run; D1 is only planned; D2 remains sealed.
D0 cases	14 cases; fixture SHA-256 f403d59e...; namespace benchmark:d0	Fictional, labelled cases used only for engineering tests.
D0 output	3 repeats; frozen 1 September 2026; output SHA-256 dcc2fbff...	The frozen result shows how the mechanism behaves.
Evidence not collected	C0 and C3: zero observations; live company/model/web: not run	There is no evidence yet about people, live sources, usefulness or production use.

Table G.1. Implementation and evidence record at 28 August 2026

Source note: Local commands, repository checksums, database update output and D0 evaluator output recorded by the author. The D0 output is preserved as `audit/d0_evaluation_2026-09-01.json`; shortened checksums are shown in full in the chapter and repository records. This information helps repeat the work, but it does not prove that the sources are true, that the findings apply elsewhere, or that the system benefits users.

Check	Result	Evidence from the run	Limit
Full automated suite	PASS	286 passed; 22 warnings; 85.58% statement coverage	Fictional, fake and local inputs; coverage does not prove that the software is complete or free from faults.
Ruff lint	PASS	No lint finding across src and tests	Only the configured static rules were checked.
Ruff format check	FAIL	Five files would be reformatted: two source and three test modules	Dated result only; later code changes were not evaluated by this record.
Strict MyPy	PASS	No issues in 46 source files	Only typed code and the configured rules were checked.
Database migration	PASS	Fresh SQLite upgrade 0001–0009; snapshot head 0009; no pending operations	Local SQLite only; production concurrency and recovery were not tested.
Diff whitespace check	PASS	git diff –check returned success	This does not check meaning or untracked files.
External/live actions	NOT RUN	No company, model, registry or public-web call	Live availability, coverage, latency and cost remain unknown.

Table G.2. Engineering check results recorded on 28 August 2026

Source note: Fresh local commands were run on the dated version recorded in EVAL-T1. PASS applies only to the named check. The formatting failure and warnings are reported because they are part of the result.

Supplementary Exhibits and Text Alternatives

Stage	Decision and action	Required local record	Reject or hold
1. Search from the problem	Search from the reporting problem, research questions and clear evidence gaps. Give each possible source a planned purpose.	Possible title, origin and planned use. Finding a source does not make it acceptable.	Sources outside scope, duplicates and search snippets are not accepted.
2. Complete local copy	Obtain a complete, readable local PDF or a permitted write-once capture before using it for a claim.	Stable local path plus origin and capture details where applicable.	Inaccessible, partial, snippet-only sources, or sources with no stable copy, are held or rejected.
3. Link the source records	Link one citation key to its reference details, origin, local path, page count and SHA-256 checksum.	The source list, checksum list and bibliography agree. Web-capture records also agree when used.	A missing or repeated key, missing row or hash, reused path, or any mismatch blocks the source.
4. Check each claim and page	Read the local full text for the exact claim and record the supporting page.	A claim-ledger row with the source purpose, pages and check result.	Claims without support, outside the source's scope or with no exact location are rejected. A citation alone does not prove truth.
5. Keep conflicts and limits	Keep conflicts, unknowns, source strength, limits of reconstructed data and fields that do not apply.	Notes about limits and conflicts stay linked to the claim and source purpose.	A hidden conflict, missing warning, false equality between sources or claim that goes too far must be repaired.
6. Final source check	Run the strict local check, then fix the cited files and their links for the section.	Readable files; matching manifest, checksums, bibliography and capture records; resolvable claim locators.	A missing or unreadable file, unresolved citation, changeable-only reference or any mismatch blocks the final check.

Table H.1. How sources were found, saved and checked

Method status: A targeted review based on the project problem. It is not systematic, exhaustive or a PRISMA review.
Source note: Developed by the author from the local source rules, not from results. Local copies, checksums and citations make sources traceable but do not prove that they are true.

Evidence class	Fact shown by the source	Careful report statement	Claim not supported	Key local literature
Crowdfunding / funding round	Attributed platform or publisher record of a campaign or round, with its date, disclosed stage or amount, and source scope where stated.	“The named source records this entity’s campaign or funding event on the stated date.” Include the exact locator and qualify platform or sample scope.	Proof of firm quality, universal investor preference, causal impact, future performance or investment merit.	estrin2024digital wasti2024 successive
Public grant or state-support record	Named programme, recipient, award or support event, date, publisher and explicit amount/currency where available.	“The publisher records the named award or support event for this legal entity.” State the date, locator, cut-off and any amount-coverage limit.	That support caused innovation or growth; that the recipient is superior; complete support history or measured impact.	mondal2025state thorne2026 funding fca2024 promotions
Companies House identity / filing event	Exact company number plus a legal name, status, incorporation or filing event from a dated register snapshot.	“At the stated cutoff, the register records this legal fact or filing for company number X.” Retain filing period and register scope.	Incorporation as active trading; complete operations, ownership or financial health; private funding, valuation or current state beyond the cutoff.	galanakis2026chrt hardman2023 small krasikov2020 ready
Reconstructed funding-lifecycle link	Conservatively linked events whose component sources, identifiers, dates, linkage rule and unmatched gaps remain visible.	“These attributed records form a partial reconstructed timeline for the same legal entity under the stated linkage rule.”	Exhaustive funding history; absence as proof of no event; a causal pathway, universal lifecycle or forecast of success.	thorne2026 funding mondal2025state

Table H.2. What early-stage company evidence can and cannot support

Meaning: Developed by the author. This is not a result, score, ranking, valuation, recommendation or cause-and-effect model.

Source note: The local sources are estrin2024digital, wasti2024successive, mondal2025state, thorne2026funding, galanakis2026chrt, hardman2023small, krasikov2020ready and fca2024promotions. Supporting pages are in claim-ledger rows 2.2-P1–2.2-P4.

Data control	Required distinction	Unsafe combination	Effect on reporting or evaluation	Key local literature
Legal identity	Keep an exact source-scoped legal identifier separate from names, domains and aliases; hold unresolved candidates or conflicts.	Name-only or fuzzy auto-merge; treating incorporation as economic activity.	Evidence can attach to the wrong company, contaminating reports and identity accuracy or coverage measures.	thorne2026funding surak2026gateways galanakis2026chrt
Observed value / missing state	Separate a value from the reason it is missing. The project distinguishes observed, zero, blank, explicit none, not applicable, not reported, not found publicly and invalid.	Treating null, blank, no record, invalid and zero as equivalent.	False totals or absence claims follow; blank-zero and missing-state evaluation becomes invalid.	gebru2021datasheets bradley2024synfintabs nikiforova2020quality
Publication / effective / retrieval / cutoff time	Retain value period, publication or effective availability, retrieval time and reporting cutoff separately.	One timestamp; retrieval treated as publication; later evidence used at an earlier cutoff.	Leakage across time, old or future claims and an unfair historical evaluation. Evidence from the wrong date must be held.	hardman2023small galanakis2026chrt kapoor2023leakage
Correction / version / snapshot	Keep write-once source bytes and checksum, data meaning, schema and source versions, and each later correction as a new dated version.	Overwrite the earlier snapshot or silently backfill a correction into a past report.	Claims and calculation totals cannot be repeated. Historical comparisons and links to approval become invalid.	gebru2021datasheets hardman2023small kapoor2023leakage
Use-case data quality	Assess explicit dimensions and executable rules against the named data object and reporting task.	One context-free quality score, or assuming open availability means fitness.	The cause of failure becomes unclear, eligibility rules are applied wrongly and results across tasks cannot be compared fairly.	nikiforova2020quality krasikov2020ready

Table H.3. Data meanings that must remain separate

Meaning: Developed by the author and not a result. The exact list of missing states is a project choice, not a published or universal standard.

Source note: The local sources are thorne2026funding, surak2026gateways, galanakis2026chrt, gebru2021datasheets, bradley2024synfintabs, nikiforova2020quality, hardman2023small, kapoor2023leakage and krasikov2020ready. Supporting pages are in claim-ledger rows 2.3-P1–2.3-P5.

Area	What must be checked again	Current evidence	What cannot be concluded
Data meanings	Measures, periods, missing states, whether sources can be found, currencies and cut-off dates	Versioned local rules; fourteen D0 cases	Accuracy for new companies, documents, sectors or periods.
Identity and sources	Official identifiers, licences, availability, source levels and exact locations	Offline replay and controlled fake web sources	Current official truth, live coverage or stable access.
Model and search	Provider, model and prompt versions; ranking; budgets; retention; and failure behaviour	Protected route built but not run live	Future model or search quality, delay or cost.
Human setting	Intended reviewer, permission, task, interface accessibility, workload and effects on decisions	Named control mechanisms; zero C0/C3 observations	Time saving, error reduction, usefulness or decision quality.
Operation and security	Sign-in, separation between users, storage, network access, monitoring, backup, incidents and recovery	Local single-user controls and controlled hostile-input tests	Hosted, multi-user, reliable or production-secure operation.
Decision use	Permitted purpose, communication review and professional accountability	No recommendation, scoring or valuation functions	Investment merit, prediction, advice or replacement of professional judgement.

Table H.4. Conditions for use in another setting

Source note: Author summary of the tested version and the remaining conditions. The table lists what must be checked before using the system elsewhere. It is not a deployment checklist, a compliance statement or proof that the move would work.

Question	Expectation from the literature	Observed evidence	Comparison
RQ1	Clear source records, data-quality rules and repeat controls make decisions easier to inspect, but do not prove that a source is true or that the system is accurate in real use.	Identity, time, source, verification and export controls ran; fourteen D0 cases repeated exactly.	Compatible in direction: this only shows that the controlled mechanism can run. It does not conflict with the reviewed literature, but there is no real-world accuracy estimate.
RQ2	Claim correctness, citation support and attribution should be assessed separately; a support gate should withhold represented unsupported material.	C2 removed six D0 false positives, retained all five supported cases and changed precision from 0.455 to 1.000 without D0 recall loss.	Compatible in how it works: the direction matches the design drawn from the literature, but the perfect values reflect the designed cases and cannot show a normal population effect.
Future human pilot	Human control may improve reflection and accountability, while also creating reliance, preference, workload and delay trade-offs.	Named review controls exist, but C0 and C3 each contain zero observations and every human outcome is null.	No observed comparison: the pilot design is prospective and no human effect is claimed.
Generalisation	Independent held-back evidence, protection against test leakage, complete records and clear uncertainty are needed before making wider claims.	D0 labels and rules were co-developed; D1 is protocol-only, D2 is sealed, and live/company comparisons are unrun.	Expected evidence gap: this agrees with the cautions in the methods literature, but it cannot support a statistical judgement that the result is anomalous or non-anomalous.

Table H.5. How the results compare with the literature (qualitative, not a non-anomaly test)

Source note: Author comparison of the Chapter 5 results with Gao et al. (2023a, 2023b), Pineau et al. (2021), Nikiforova et al. (2020), Gebru et al. (2021), Kapoor & Narayanan (2023), Amershi et al. (2019) and Bućinca et al. (2021). The sources motivate expectations and limits of the method. None provides a population distribution that can be compared directly with D0.

Risk area	Current control	Current evidence	What is still required
Data and privacy	Classification, purpose declaration, local restricted/internal boundary, public-text redaction	Policies, schemas and controlled tests	Approved lawful-purpose, retention, deletion, backup and incident responsibilities.
External model	Public or fictional data only, reviewed route, strict schema, budgets and store=False	Code and fake-provider tests; five bounded synthetic smoke runs, 26 August 2026 (Section 7.2)	Eligible account settings, retention and deletion evidence, and a current provider review.
Untrusted content	URL candidates only, protected collection, instruction filtering and exact-passage checks	Controlled malformed-content and grounding cases	Independent hostile-input testing and continuing maintenance of source rules.
Network transport	HTTPS, DNS, redirect, port, file type and size checks with public-IP pinning	Controlled SSRF, rebinding and redirect cases	Separate production network access, sign-in and separation between users.
Resource and recovery	Budgets, retries, cancellation, fixed fingerprints, checksums and staged export	Task and export recovery tests	Backup, service targets, monitoring, recovery after a forced stop, and incident exercises.
Decision and persons	No recommendation, valuation, suitability, profiling or reputation scoring	Requirements and output contracts	Separate authority for any changed purpose, audience or regulated decision use.
Research validity	Fixed rules, explicit missing results, source and dataset lists, and D0 repeats	Fourteen designed cases; D1/D2 and C0/C3 absent	Independent references, a sealed future evaluation and authorised human observations.
Operation and access	Local single-user deployment and meaningful printable output	Local tests and visual PDF inspection	Full accessibility audit, assistive-technology study and evaluation of a production design.

Table H.6. Governance controls, evidence and remaining requirements

Source note: Author summary of the implemented controls, Chapter 5 evidence and declared holds. This table records the dissertation’s evaluated state; it is not legal advice, a compliance assessment, a penetration test or deployment approval.

Stage	Activity requiring permission	Evidence needed to proceed	Result unavailable before this stage
1	Authorised matched business pilot	Fixed companies, periods, sources, reference decisions, completion rule, staff owners and acceptance criteria	Supported-claim quality, human time, edits, source failures, completion and cost per report.
2	Live check using only counts, checksums and locations; no report content	Named permission, approved routes, account and retention evidence, and a write-once run record	Evidence about live connection, retention route and basic operation.
3	Official connectors and a fixed public-company source benchmark	Fixed companies, cut-off date, sources, queries, budgets and independent passage reference	Estimates for added supported discovery, failures, resources and cost.
4	One sealed final evaluation	Unchanged study plan, no tuning on held-back cases and analysis that includes failures	Claim-quality and reliability estimates for new cases.
5	Managed-platform comparison	Equivalent inputs, source windows, output contract, scoring rules and recorded setup effort	Comparative setup, maintenance, traceability, failure recovery and total cost.
6	Accessibility, security and operational assessment	Assistive-technology audit, independent security scope, ownership, monitoring and recovery exercises	Any justified usability, resilience, security or production-readiness conclusion.

Table H.7. Ordered evidence plan after this dissertation

Source note: Author summary of missing evidence and stopping conditions. Each stage requires separate permission and a fixed study plan. The order does not imply approval, success or an obligation to deploy.

Generative-AI Use Disclosure

Acknowledgement and permitted purpose

OpenAI Codex (<https://openai.com/codex/>) was used as a support tool during this project. Its use was limited to five areas: planning and refining the report structure; creating and revising visualisations; supporting research and source-checking; providing feedback on clarity, flow and academic presentation; and assisting with technical implementation, tests, LaTeX changes and validation commands. It was not treated as an academic source, an empirical participant or evidence that an implementation or evaluation had succeeded.

The candidate retained responsibility for the submitted work. Proposed text, code and visual changes were compared with the current repository, saved evaluation outputs and locally held sources. Suggestions that could not be verified were rejected, narrowed or recorded as missing evidence. Academic claims were retained only where the cited source had been checked, and generated technical changes were accepted only after the relevant checks were run.

Material prompts used for the final supervisor-feedback revision

The material prompts for the final revision asked the tool to:

1. review the existing dissertation against the supervisor's feedback, focusing on what was actually validated; the business problem, users and reporting process; the need for role separation rather than a simpler system; literature and technical flow; repeated limitations; appendix placement; accepted and rejected claims; the business pilot; plain language; direct research-question answers; and captions below tables and figures;
2. update the remediation plan to use a 15,000-word main-body target with a ten-per-cent tolerance, tick the seven completed pro-forma content items while leaving ethics approval pending, consider whether bounded implementation and evaluation could repair an overclaim, and apply the supplied WMG generative-AI acknowledgement guidance; and
3. implement the remaining evidence, title, result, caption, Abstract, word-count, exhibit, review-status and PDF findings, then perform a final dissertation-reviewer gate.

How the outputs were modified and checked

Report-structure suggestions were reconciled with the active research questions and the 13,500–16,500-word main-body range. Visual outputs were kept only when they supported a named part of the argument, used the shared print-safe style and remained legible in the compiled PDF. Research suggestions were checked against the local source library and were not cited as AI output. Feedback was applied selectively in British English, with repeated limitations consolidated and technical detail retained only where it supported the research argument. Technical and LaTeX changes were inspected against the current implementation, tested with the repository’s focused checks, and then compiled and visually reviewed. The prompt list and disclosure scope follow the candidate-supplied WMG section 3.2 guidance preserved in `sources/WMG_AI_USE_GUIDANCE_CANDIDATE_SUPPLIED.md`.